GREEN CAMPUS WASTE AND RECYCLING TRACKER

A PROJECT REPORT

Submitted by

SRI JANANI B

(241401108)

in partial fulfilment of the End Semester Examination for the course

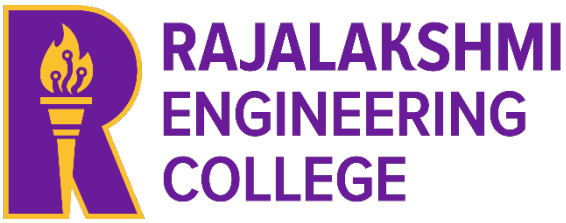
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

COMPUTER SCIENCE AND BUSINESS SYSTEMS



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

NOVEMBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report titled "GREEN CAMPUS WASTE AND RECYCLING TRACKER" is the bonafide work of SRI JANANI B (241401108), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

SIGNATURE

Dr. K.R. Sowmia
HEAD OF THE DEPARTMENT
Department of Computer Science and Business Systems
Rajalakshmi Engineering College
Thandalam – 602 105

SIGNATURE

Mr. M.L. Alphin Ezhil Manuel
COURSE FACULTY
Assistant Professor
Department of Computer Science and Business Systems
Rajalakshmi Engineering College

Submitted for the End Semester Examination (Individual Project) held on 03 November 2026.

INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, SRI JANANI B (241401108), hereby declare that the project report titled “GREEN CAMPUS WASTE AND RECYCLING TRACKER”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 03 November 2026

SIGNATURE OF THE STUDENT
SRI JANANI B
241401108

ACKNOWLEDGEMENT

I express my sincere gratitude to our Chairman Mr. S. Meganathan, B.E., F.I.E., Chairperson Dr. Thangam Meganathan, Ph.D., and Vice-Chairman Mr. Abhay Shankar Meganathan, B.E., M.S., for providing the infrastructure and facilities that made this work possible.

I am grateful to our Principal, Dr. S.N. Murugesan, M.E., Ph.D., Rajalakshmi Engineering College, for the constant encouragement extended to students to take up industry-relevant learning.

I thank Dr. K.R. Sowmia, Ph.D., Associate Professor and Head of the Department, Department of Computer Science and Business Systems, for the guidance and support provided throughout the course.

I owe my sincere thanks to my course faculty, Mr. M.L. Alphin Ezhil Manuel, Assistant Professor, for the valuable guidance, reviews and feedback that shaped this application and report.

I also acknowledge Zoho Corporation and the Zoho Creator Education Programme for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

SRI JANANI B

ABSTRACT

Educational campuses generate large volumes of mixed waste every day from classrooms, laboratories, hostels and canteens. At Rajalakshmi Engineering College, as in most campuses, collection is recorded in paper registers or loose spreadsheets, vendor pickups are arranged by phone, and hazardous laboratory and electronic waste is handled without a traceable record. As a result the sustainability team cannot see how much waste is produced, where it comes from, how much is actually recycled or how much carbon is saved, and monthly reports must be compiled by hand.

This project develops the Green Campus Waste and Recycling Tracker on Zoho Creator, a low-code platform chosen because it provides forms, a managed database, reports, dashboards, workflow automation, role-based security and mobile access without building and hosting a full software stack. The application is deliberately kept simple: a single Waste Entry form organised into six sections, a list report and a kanban waste board grouped by category, and one Green Campus Dashboard page with six charts. Ten Deluge workflows fill default values, show or hide fields, calculate CO2 saved, Green Points and the estimated vendor value, validate weight, date, hazardous-waste and pickup rules, enforce role checks and send confirmation, alert and approval e-mails. An approved pickup is pushed to Zoho Calendar through an OAuth connection and the Calendar REST API. Four permission profiles (Sustainability Officer, Department Coordinator, Waste Collector and Student Volunteer) restrict actions and hide personal data.

The application was tested with valid and invalid entries and with 41 sample records; errors found during development, such as a TIME-value concatenation error, an unverified sender address and an empty date-time conversion, were traced with info logs and fixed. The tracker replaces manual registers with one consistent record per collection, gives instant category, location and disposal summaries, and makes every pickup approval traceable.

Keywords: *Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Role-Based Access Control; Campus Waste Management*

PROJECT AT A GLANCE

Attribute	Details
Project Title	Green Campus Waste and Recycling Tracker
Application Domain	Environment and Sustainability / Campus Facilities Management
Student Name & Register No.	SRI JANANI B – 241401108
Zoho Creator Edition	Zoho Creator (trial edition)
Application Name (Link Name)	Green_Campus_Waste_Recycling
Published / Live URL	https://creatorapp.zoho.in/srijanani.b.2024.csbs_raj272/environment/development/green-campus-waste-recycling/#Form:Waste_Entry
Demo Credentials (per role)	Admin: srijanani.b.2024.csbs@rajalakshmi.edu.in (password provided separately). Waste Collector and Student Volunteer access is granted by the administrator through Share; role-wise test logins are provided to the examiner on request.
Number of Forms / Subforms	1 form / 0 subforms
Number of Reports (by type)	List: 1 (Waste Entry Report) Kanban: 1 (waste_board) Calendar: 0 Chart: 6 charts on the dashboard page Others: 0
Number of Pages / Dashboards	1 (Green Campus Dashboard)
Number of Workflows	Form: 10 Schedule: 0 Approval: 0 Blueprint: 0 Batch: 0
Deluge Functions / Custom Actions	10 Deluge scripts (one per form workflow); 0 standalone functions
Integrations	Zoho Calendar (REST API through connection zoho_calendar_connection); Zoho Mail (sendmail notifications)
Roles & Profiles Defined	Sustainability Officer (Administrator), Department Coordinator, Waste Collector, Student Volunteer
Environments Used	Development (trial account), promoted to Stage and Production as version 1.0
Date of Production Deployment	01 November 2026 (version 1.0 in Stage and Production)

TABLE OF CONTENTS

ABSTRACT....iv
PROJECT AT A GLANCE....v
LIST OF TABLES....viii
LIST OF FIGURES....ix
LIST OF ABBREVIATIONS....x
INTRODUCTION....1
1.1 Background....1
1.2 Problem Statement....1
1.3 Objectives....2
1.4 Scope of the Project....2
1.5 Justification for a Low-Code Approach....3
1.6 Organisation of the Report....3
LITERATURE SURVEY AND SYSTEM STUDY....4
2.1 Study of the Existing System....4
2.2 Limitations of the Existing System....5
2.3 Low-Code versus Traditional Development....5
2.4 Literature Survey....6
2.5 Emerging Trends in Low-Code Development....7
2.6 Proposed System....7
REQUIREMENT ANALYSIS....8
3.1 Stakeholders and User Roles....8
3.2 Functional Requirements....8
3.3 Non-Functional Requirements....10
3.4 Hardware and Software Requirements....10
3.5 Use Case Model....10
SYSTEM DESIGN....12
4.1 System Architecture....12
4.2 Data Model....12
4.3 Form Design....14
4.4 Process and Workflow Design....15
4.5 User Interface and Navigation Design....17
4.6 Access Control Design....17
IMPLEMENTATION....18
5.1 Application Setup....18
5.2 Forms and Data Capture....18
5.3 Reports....21
5.4 Pages and Dashboards....23
5.5 Deluge Scripting....24
5.6 Workflows and Automation....26
5.7 Integrations....27

5.8 User Management and Role-Based Access....29
5.9 AI and Analytics Features....31
5.10 Pro-Code Extension....32
5.11 Mobile and Portal Access....32
TESTING AND DEBUGGING....33
6.1 Testing Strategy....33
6.2 Test Cases....33
6.3 Debugging Deluge Scripts....35
6.4 Defect Log....35
6.5 User Acceptance Testing....36
DEPLOYMENT, SECURITY AND GOVERNANCE....37
7.1 Environments and Deployment Strategy....37
7.2 Performance Optimisation....39
7.3 Security Features....40
7.4 Governance and Administration....40
RESULTS AND DISCUSSION....42
8.1 Application Output....42
8.2 Achievement of Objectives....45
8.3 Comparison with the Existing System....46
8.4 Limitations of the Low-Code Approach....46
8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum....46
CONCLUSION AND FUTURE ENHANCEMENTS....47
9.1 Conclusion....47
9.2 Future Enhancements....47
9.3 Learning Outcomes and Career Relevance....47
REFERENCES....48
APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST....49
APPENDIX B – COURSE OUTCOME MAPPING....52
APPENDIX C – COMPLETE DELUGE SOURCE CODE....53
APPENDIX D – USER MANUAL....62
D.1 Accessing the Application....62
D.2 Instructions for Department Coordinators, Waste Collectors and Student Volunteers....62
D.3 Instructions for the Sustainability Officer (Administrator)....62

LIST OF TABLES

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application..... 5
Table 2.2: Summary of Literature Survey..... 6
Table 3.1: Stakeholders and User Roles..... 8
Table 3.2: Functional Requirements..... 8
Table 3.3: Non-Functional Requirements..... 10
Table 3.4: Hardware and Software Requirements..... 10
Table 3.5: Use Case Description – Record a Waste Entry and Request a Pickup..... 11
Table 4.1: Entities and Relationships..... 13
Table 4.2: Field Specification – Waste Entry Form..... 14
Table 4.3: Role–Permission Matrix..... 17
Table 5.1: Reports Implemented..... 21
Table 5.2: Deluge Scripts Implemented..... 24
Table 5.3: Workflows Implemented..... 26
Table 5.4: Integrations Implemented..... 27
Table 6.1: Test Cases..... 33
Table 6.2: Defect Log..... 35
Table 6.3: User Acceptance Feedback..... 36
Table 7.1: Deployment Checklist..... 38
Table 7.2: Performance Optimisation Results..... 40
Table 7.3: Security Controls Implemented..... 40
Table 8.1: Mapping of Objectives to Outcomes..... 45
Table 8.2: Existing System versus Proposed Application..... 46
Table A.1: Feature Coverage Checklist..... 49
Table B.1: Mapping of Course Outcomes to Report Chapters..... 52

LIST OF FIGURES

Figure 2.1: As-Is Process Flow of the Existing Waste Collection System.....4
Figure 3.1: Use Case Diagram of the Green Campus Waste and Recycling Tracker. 11
Figure 4.1: System Architecture of the Green Campus Waste and Recycling Tracker 12
Figure 4.2: Entity–Relationship Diagram of the Application.....13
Figure 4.3: Workflow / Process Flow for Recording a Waste Entry and Approving a Pickup..... 16
Figure 4.4: Navigation Structure and Wireframe of the Green Campus Dashboard. .17
Figure 5.1: Application Home and Navigation Sections.....18
Figure 5.2: Waste Entry Form in the Form Builder.....19
Figure 5.3: Waste Entry Form in Live Mode.....20
Figure 5.4: Waste Entry Report (List) and waste_board Kanban Report.....23
Figure 5.5: Green Campus Dashboard Page.....24
Figure 5.6: Form Workflows Configured on the Waste Entry Form.....27
Figure 5.7: Zoho Calendar Connection and the Created Pickup Event.....28
Figure 5.8: Permission Profiles and Role Hierarchy.....30
Figure 5.9: Category Filter Applied on the Waste by Category Chart.....32
Figure 5.10: Phone Layout of the Waste Entry Form (Form Customization – Phone) 32
Figure 6.1: Workflow Log Showing Deluge info Output.....35
Figure 7.1: Environments and Publish Settings.....38
Figure 7.2: Record History and Administration Settings.....41
Figure 8.1: Scenario Step 1 – Coordinator Records Plastic Waste and Requests a Pickup..... 42
Figure 8.2: Scenario Step 2 – Confirmation and Alert E-mails.....44
Figure 8.3: Scenario Step 3 – Approval and Zoho Calendar Event.....45

LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CO2	Carbon Dioxide
CRUD	Create, Read, Update, Delete
CSV	Comma-Separated Values
DS	Deluge Script (Zoho Creator application definition file)
ER	Entity–Relationship
JSON	JavaScript Object Notation
LCAP	Low-Code Application Platform
OAuth	Open Authorization
PII	Personally Identifiable Information
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho's AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Rajalakshmi Engineering College is a large residential campus with academic blocks, laboratories, a central library, separate boys' and girls' hostels, a main canteen, sports facilities and administrative offices. Several thousand students and staff use these spaces every day, and each of them produces a different kind of waste: paper and cardboard from classrooms and offices, plastic and food waste from the hostels and canteen, glass and chemical containers from the laboratories, and electronic waste such as old keyboards, monitors and batteries from the computer and electronics labs.

Waste is currently handled by housekeeping and collection staff who empty bins on each shift, a supervisor who notes approximate quantities in a register, and a small sustainability team that arranges recyclable material to be picked up by outside vendors. India's Solid Waste Management Rules, 2016 and E-Waste (Management) Rules, 2022 expect institutions to segregate waste at source and to hand recyclable and hazardous streams to authorised handlers, so the college needs reliable records of what is generated and where it goes [5], [6].

This project builds a digital tracker for that process on Zoho Creator. Every collection is entered once on a structured form; the application calculates its environmental impact, routes pickup requests for approval, notifies the people involved and shows the whole campus picture on a single dashboard.

1.2 Problem Statement

The college has no single, reliable record of the waste it collects. Quantities are written in registers or scattered spreadsheets, vendor pickups are arranged informally, and the sustainability team cannot see in real time how much waste each location produces, how it is disposed of or how much of it is recycled.

- Collection data is recorded on paper or in separate spreadsheets, so entries are duplicated, incomplete or lost, and totals have to be counted by hand.
- There is no check at the time of entry: wrong weights, missing details and hazardous waste sent to landfill are not caught.

- Vendor pickups are requested by phone or messaging, with no approval step, no schedule and no record of the expected value.
- Management has no real-time view of waste by category, location or disposal method, or of the CO2 saved by recycling.
- Anyone with the spreadsheet can view or change any entry, including the contact details of students and staff.

1.3 Objectives

- To design a structured data model and a single, sectioned Waste Entry form that captures every collection with mandatory fields and consistent drop-down values.
- To automate the calculation of CO2 saved, Green Points and estimated vendor value, and to enforce waste-handling rules, using Deluge scripts in form workflows.
- To provide a list report, a kanban waste board and one dashboard page with six charts for campus-wide monitoring.
- To integrate the application with Zoho Calendar through a connection and its REST API so that every approved pickup is scheduled automatically, and to notify users through e-mail.
- To configure permission profiles and role checks so that each user can perform only the actions relevant to their role and personal data is protected.
- To test the application with valid and invalid data, record and fix defects, and prepare it for deployment.

1.4 Scope of the Project

In scope: recording waste collections (date, shift, location, department, category, weight, disposal method and segregation); hazardous-waste notes; vendor pickup requests with an estimated value; automatic CO2 and Green Points calculation; status tracking from Logged to Recycled; e-mail notifications; Zoho Calendar scheduling of approved pickups; a list report, a kanban board and a dashboard; four user roles (Sustainability Officer, Department Coordinator, Waste Collector and Student Volunteer); access through the web and the Zoho Creator mobile app.

Out of scope: weighing-scale or IoT sensor integration, vendor billing and payments, a public customer portal, route optimisation for collection vehicles, and certified carbon

accounting. The CO2 factors used are indicative values configured in the application and can be updated by the administrator.

1.5 Justification for a Low-Code Approach

The tracker is a typical departmental business application: a few forms, clear rules, notifications, reports and role-based access, used by non-technical staff. Building it with traditional code would need a database, a back-end, a web front-end, an e-mail service, hosting and separate mobile support. Zoho Creator provides all of these as managed services; the form, report and page builders are visual, and Deluge, Zoho's scripting language, is used only where custom logic is needed [2], [3].

This approach shortens development from weeks to days, lets a citizen developer such as a member of the sustainability team maintain the application, and provides built-in hosting, mobile apps, connections to other Zoho services and role-based security. Low-code platforms are widely recognised for exactly this class of data-centric business application [7], [8].

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The current process is manual. Housekeeping staff empty the bins in each building on the morning, afternoon or evening shift. The quantity is weighed only occasionally; usually the supervisor estimates it and writes the location, type and approximate weight in a register. At the end of the week the entries are copied into a spreadsheet.

When recyclable material accumulates, the supervisor phones or messages a known vendor, who collects it and gives a paper receipt. Hazardous laboratory waste and e-waste are stored separately and handed over when someone remembers to arrange it. At the end of the month the sustainability team compiles totals by hand for the management. Figure 2.1 shows this as-is process.

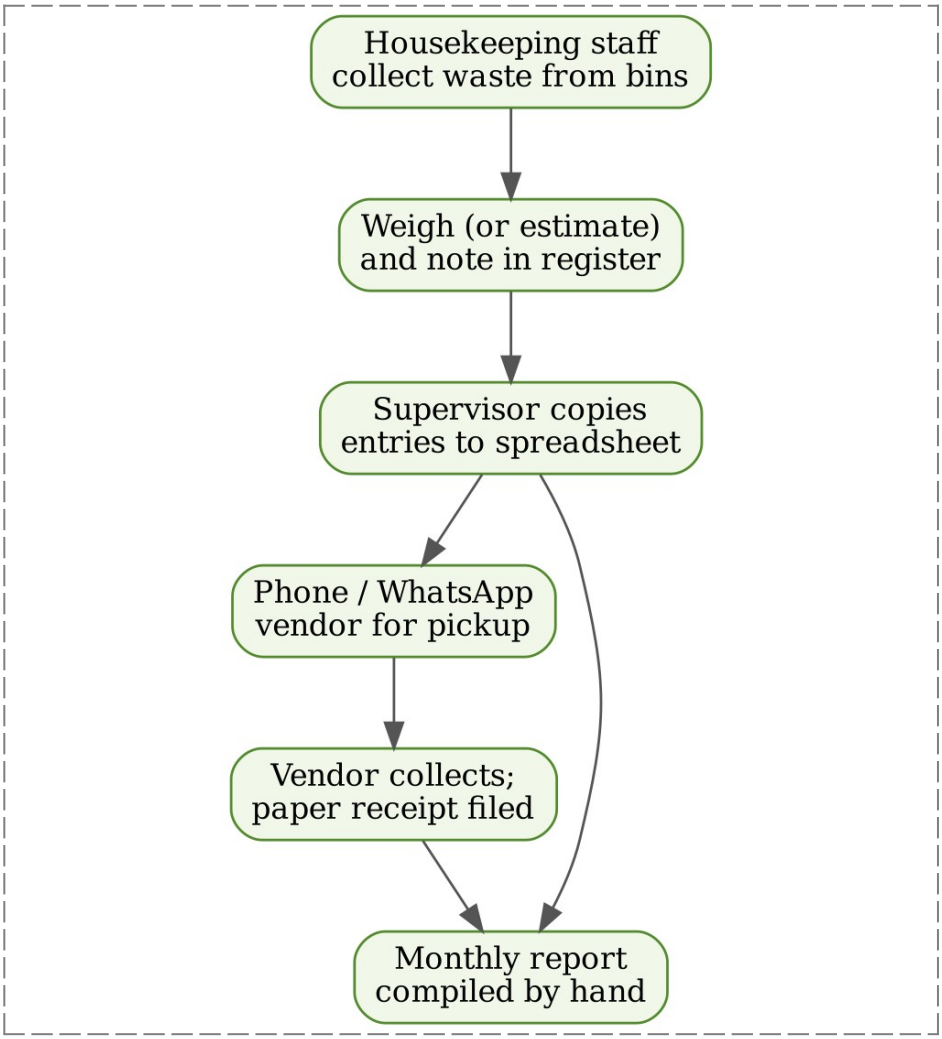


Figure 2.1: As-Is Process Flow of the Existing Waste Collection System

2.2 Limitations of the Existing System

- Data is entered twice (register, then spreadsheet), which wastes time and introduces copying errors.
- There is no validation, so impossible weights, missing details and hazardous waste sent to landfill go unnoticed.
- Pickups have no approval step or schedule, and the expected value of recyclables is never recorded.
- Summaries by category, location and disposal method are produced only once a month, by hand.
- There is no access control or audit trail; personal contact details are visible to everyone.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	6–8 weeks for database, back-end, front-end and deployment	About 1 week for the form, reports, dashboard and workflows
Skills required	Full-stack skills: SQL, a back-end language, HTML/CSS/JavaScript, DevOps	Zoho Creator builders and basic Deluge scripting
Hosting & infrastructure	Server or cloud hosting, database administration, backups	Hosted and backed up by Zoho
Mobile application	Separate responsive or native app development	Automatic Zoho Creator mobile app for the same application
Security & access control	Login, sessions and role checks coded by hand	Built-in users, roles, permission profiles and PII settings
Maintenance & change	Code changes, testing and redeployment	Edit fields, reports and workflows visually; changes are live immediately
Customisation limits	Unlimited	Limited to platform features; complex UI needs widgets or pro-code
Cost	Developer time plus hosting	Per-user subscription (trial edition used for this project)

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Sahay, Indamutsa, Di Ruscio and Pierantonio, IEEE SEAA, 2020	Supporting the understanding and comparison of low-code development platforms	Compares major low-code platforms and their common features: visual modelling, data management, workflow and integration.	Guided the choice of Zoho Creator features used for forms, workflows and integration.
2	Bock and Frank, Business & Information Systems Engineering, 2021	Low-Code Platform	Defines low-code platforms and their typical use for departmental business applications by citizen developers.	Supports the justification for a low-code approach for a campus team.
3	Di Ruscio, Kolovos, de Lara, Pierantonio, Tisi and Wimmer, Software and Systems Modeling, 2022	Low-code development and model-driven engineering: Two sides of the same coin?	Relates low-code platforms to model-driven engineering and discusses their limitations.	Used in the discussion of low-code limitations and the pro-code spectrum.
4	Kaza, Yao, Bhada-Tata and Van Woerden, World Bank, 2018	What a Waste 2.0: A Global Snapshot of Solid Waste Management to 2050	Shows that reliable data on waste generation and disposal is the first step to better waste management.	Motivates recording every collection with category, weight and disposal method.
5	Ministry of Environment, Forest and Climate Change, Govt. of India, 2016 and 2022	Solid Waste Management Rules, 2016; E-Waste (Management) Rules, 2022	Require segregation at source and hand-over of recyclable, hazardous and electronic waste to authorised agencies.	Basis for the segregation field, the hazardous-waste rules and the vendor pickup flow.
6	Zoho Corporation, online documentation	Zoho Creator Help and Deluge Reference	Documents forms, reports, pages, workflows, connections, invokeurl and user permissions.	Primary reference for implementing every component of the application.

2.5 Emerging Trends in Low-Code Development

Several current trends shaped the design. AI-assisted application building and Zia features show how low-code platforms are moving towards generating forms and insights automatically; this project uses the platform's visual builders and keeps AI features as a future enhancement. Citizen development is reflected in keeping the application simple enough for a sustainability coordinator to maintain. Hyper-automation is reflected in chaining validation, calculation, notification and calendar scheduling on a single submission. Composable applications are reflected in reusing other Zoho services, Mail and Calendar, through connections instead of rebuilding them.

2.6 Proposed System

The proposed system replaces the register and spreadsheet with one Zoho Creator application. Each collection is recorded once on the Waste Entry form, which fills the date and the user's details automatically and uses fixed drop-down lists for locations, departments, categories, disposal methods and vendors, removing double entry.

Deluge scripts validate every entry, block hazardous waste from landfill and require handling notes, and calculate the CO2 saved, Green Points and pickup value. Pickup requests move through clear statuses and can be approved only by the Sustainability Officer; approval sends an e-mail and creates a Zoho Calendar event. The list report, the kanban waste board and the dashboard give instant summaries, and permission profiles restrict actions and hide personal data. Each limitation listed in Section 2.2 is therefore addressed directly.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Sustainability Officer (Administrator)	Faculty or staff member responsible for campus sustainability; owner of the application.	Manage users and roles, approve or reject pickups, close records, view all data, reports and the dashboard.
Department Coordinator	Staff member who coordinates waste for a department, hostel or canteen.	Record waste entries, request vendor pickups, view the report, the waste board and the dashboard.
Waste Collector	Housekeeping or collection staff member.	Record collections, view the waste board, update collection progress; no access to personal data.
Student Volunteer	Member of the green campus or eco club.	Record waste entries and view the dashboard; no access to other users' personal data.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow a user to record a waste entry with date, shift, location, department, category, weight, disposal method and segregation as mandatory fields.	All	High
FR-02	The system shall fill the log date, the user's name and e-mail and the status 'Logged' automatically when the form opens.	All	Medium
FR-03	The system shall calculate CO2 saved and Green Points automatically when the category, weight, disposal method or segregation changes.	All	High
FR-04	The system shall show the vendor, pickup date and estimated value fields only when a pickup is required, and calculate the estimated value from the vendor's rate.	Coordinator, Officer	High
FR-05	The system shall show the hazard notes field for hazardous waste and e-waste, and require it before saving.	All	High
FR-06	The system shall reject a weight that is zero, negative or above 1000 kg, and a log date in the future.	All	High

FR ID	Requirement	Role(s)	Priority
FR-07	The system shall prevent hazardous waste from being recorded with the disposal method 'Landfill'.	All	High
FR-08	The system shall require a vendor and a pickup date on or after today when a pickup is requested, and set the status to 'Pickup Requested'.	Coordinator, Officer	High
FR-09	The system shall allow only the Sustainability Officer to set the status to Approved, Collected, Recycled or Rejected.	Officer	High
FR-10	The system shall e-mail a confirmation with the CO2 saved and Green Points to the person who recorded the entry.	All	Medium
FR-11	The system shall e-mail the administrator when hazardous waste, 50 kg or more, or a pickup request is recorded.	Officer	Medium
FR-12	The system shall, when a pickup is approved, e-mail the requester and create an event in Zoho Calendar on the pickup date.	Officer	High
FR-13	The system shall provide a list report of all entries with search, filter and sort.	All except Student	High
FR-14	The system shall provide a kanban board grouping entries by waste category with record counts.	Collector, Coordinator, Officer	Medium
FR-15	The system shall provide a dashboard with waste by category, disposal split, daily trend, waste by location, CO2 saved by category and Green Points by department.	All	High
FR-16	The system shall hide personal data such as e-mail addresses from the Waste Collector and Student Volunteer profiles.	Officer	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports and the dashboard with up to 1,000 entries shall load within 3 seconds on a normal campus internet connection.
NFR-02	Security	Users shall access only the forms, reports and actions permitted by their profile; personal data fields are marked as PII.
NFR-03	Usability	A new user shall be able to record an entry in under one minute; the form is grouped into six labelled sections and works on web and the Zoho Creator mobile app.
NFR-04	Availability	The application shall be available whenever Zoho Creator is available, as it is hosted by Zoho.
NFR-05	Maintainability	Field link names and workflow names shall be descriptive, and value lists, CO2 factors and vendor rates shall be editable in one place.
NFR-06	Auditability	Every record shall keep its added user and time, and status changes and remarks shall be traceable through the record history.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (trial edition), web-based
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Workflow info logs in Zoho Creator (Postman optional)
Other Zoho services	Zoho Calendar (REST API through a connection), Zoho Mail (sendmail)
Mobile	Zoho Creator mobile app (Android / iOS)
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

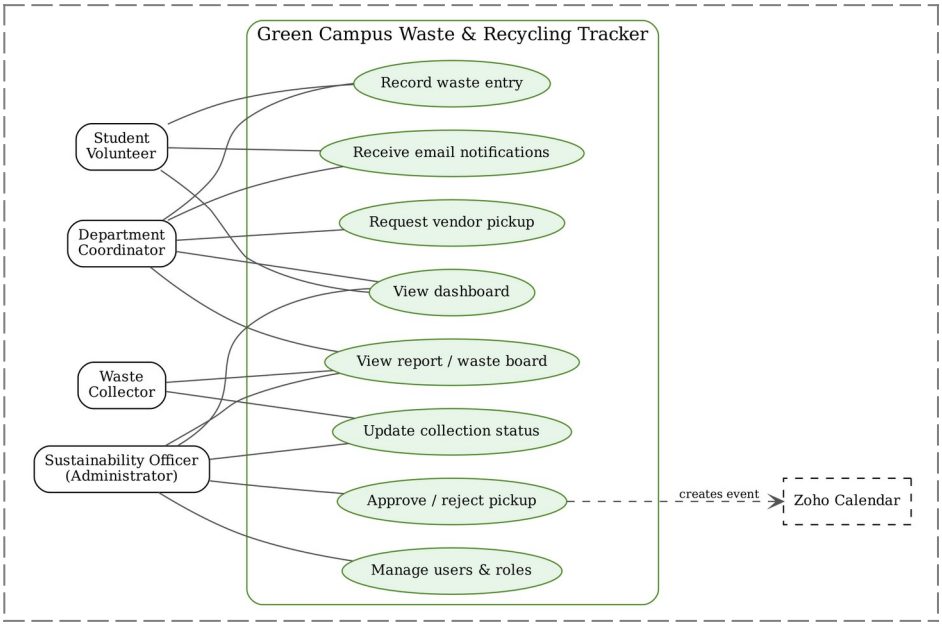


Figure 3.1: Use Case Diagram of the Green Campus Waste and Recycling Tracker

Table 3.5: Use Case Description – Record a Waste Entry and Request a Pickup

Use Case ID	UC-01
Use Case Name	Record a waste entry and request a vendor pickup
Actor(s)	Department Coordinator (primary); Sustainability Officer and Zoho Calendar (secondary)
Pre-condition	The user is logged in and has permission to add Waste Entry records.
Main Flow	1. The user opens the Waste Entry form; the date, name, e-mail and status are filled automatically. 2. The user selects the location, department, category, weight, disposal method and segregation; CO2 saved and Green Points appear. 3. The user selects 'Vendor Pickup Required = Yes', chooses a vendor and a pickup date; the estimated value appears. 4. The user submits; the record is saved with status 'Pickup Requested'. 5. A confirmation e-mail is sent to the user and an alert to the Sustainability Officer. 6. The officer sets the status to 'Approved'; the requester is e-mailed and a Zoho Calendar event is created.
Alternate Flow	3a. Weight is 0 or above 1000 kg, the date is in the future, hazardous waste has no notes or is sent to landfill, or no vendor/valid pickup date is chosen: an alert is shown and the record is not saved. 6a. A non-admin user tries to approve: the change is blocked.
Post-condition	The entry is stored with its calculated impact and status, appears in the report, board and dashboard, and the approved pickup is on the calendar.
Related FR	FR-01 to FR-12

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

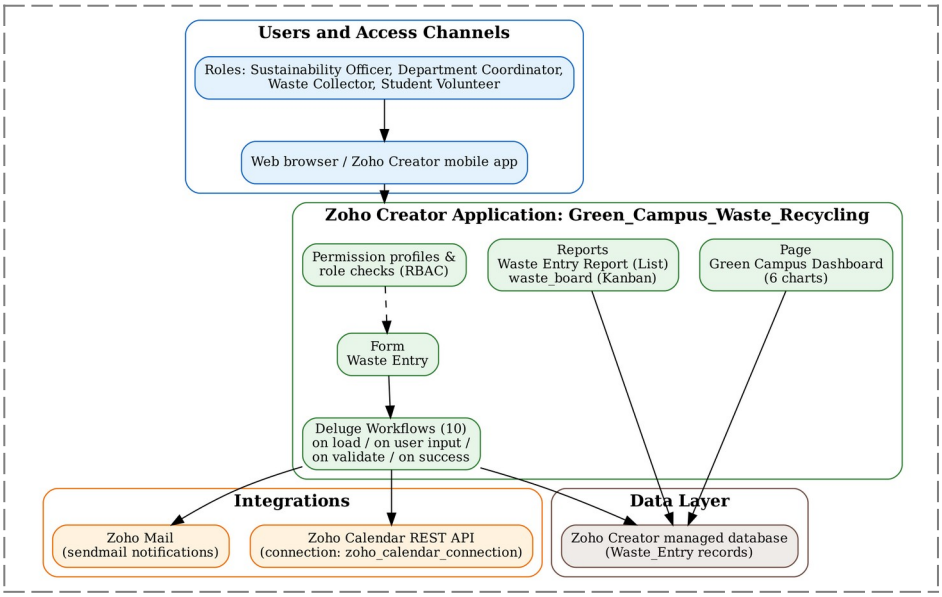


Figure 4.1: System Architecture of the Green Campus Waste and Recycling Tracker

The architecture has four layers. The user layer contains the four roles, who reach the application through a web browser or the Zoho Creator mobile app. The application layer is the Zoho Creator application Green_Campus_Waste_Recycling: the Waste Entry form for data capture, the Waste Entry Report and the waste_board kanban for viewing records, the Green Campus Dashboard page for analytics, ten Deluge form workflows for automation, and permission profiles with role checks for security.

The data layer is Zoho Creator's managed database, which stores every Waste Entry record together with system fields such as the record ID, the added user and the added time. The integration layer connects the application to other Zoho services: Zoho Mail delivers notifications sent with the sendmail task, and Zoho Calendar receives pickup events through its REST API, called with invokeurl over the OAuth connection zoho_calendar_connection.

4.2 Data Model

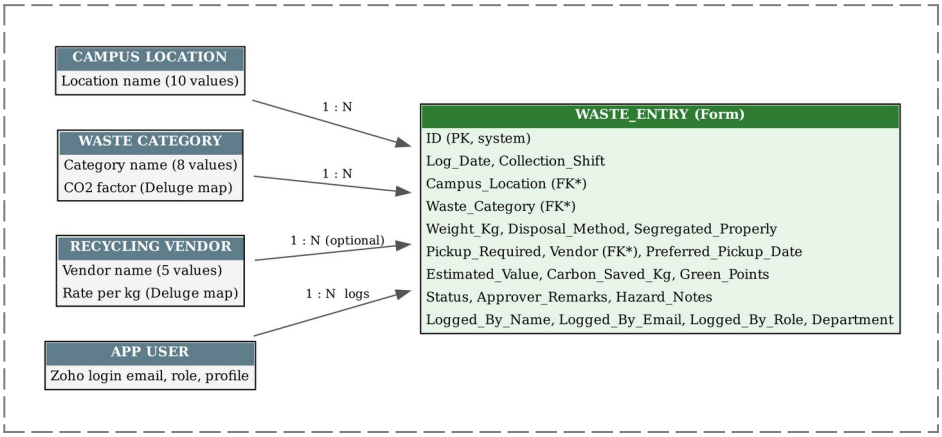


Figure 4.2: Entity–Relationship Diagram of the Application

The central entity is the Waste Entry, stored in the Waste_Entry form. Campus Location, Waste Category and Recycling Vendor are reference entities that each relate one-to-many to Waste Entry, and every entry is logged by one application user. To keep the application simple for its users, these reference entities are implemented as fixed drop-down value lists on the Waste Entry form instead of separate forms, and the CO2 factor of each category and the rate of each vendor are held in Deluge maps used by the calculation script. They can be promoted to separate master forms with lookup fields if the campus later needs to manage them as data (Section 9.2).

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Waste Entry (Waste_Entry form)	ID, Log_Date, Waste_Category, Weight_Kg, Status	Campus Location, Waste Category, Recycling Vendor, App User	Many-to-One with each	Central form; stores one record per collection
Campus Location	Location name (10 values)	Waste Entry	One-to-Many	Drop Down field Campus_Location
Waste Category	Category name (8 values), CO2 factor	Waste Entry	One-to-Many	Drop Down field Waste_Category + Deluge factor map
Recycling Vendor	Vendor name (5 values), rate per kg	Waste Entry	One-to-Many (optional)	Drop Down field Vendor + Deluge rate map

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
App User	Zoho login e-mail, role, profile	Waste Entry	One-to-Many	Logged_By_Email , Added User system field, permission profile

4.3 Form Design

Table 4.2: Field Specification – Waste Entry Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Log Date	Log_Date	Date	Yes	Default: today; not in the future	Section 1 – Entry Details
Collection Shift	Collection_Shift	Radio	Yes	Morning / Afternoon / Evening	
Campus Location	Campus_Location	Drop Down	Yes	10 campus locations	
Department	Department	Drop Down	Yes	12 departments / units	
Waste Category	Waste_Category	Drop Down	Yes	8 categories	Section 2 – Waste Details
Weight (kg)	Weight_Kg	Decimal	Yes	> 0 and ≤ 1000	
Disposal Method	Disposal_Method	Drop Down	Yes	Recycled / Composted / Sent to Vendor / Hazardous Disposal / Landfill	Landfill blocked for hazardous waste
Segregated Properly?	Segregated_Properly	Radio	Yes	Yes / No	+5 Green Points if Yes
Hazard / Handling Notes	Hazard_Notes	Multi Line	Conditional	Required for Hazardous and E-Waste	Shown only for those categories
Vendor Pickup Required?	Pickup_Required	Radio	Yes	Default: No	Section 3 – Recycling Pickup
Recycling Vendor	Vendor	Drop Down	Conditional	Required when pickup = Yes	5 vendors
Preferred Pickup Date	Preferred_Pickup_Date	Date	Conditional	On or after today	
Estimated Value (Rs)	Estimated_Value	Decimal	—	Weight × vendor rate	Calculated; read-only

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
CO2 Saved (kg)	Carbon_Saved_Kg	Decimal	—	Weight × category factor; 0 for landfill	Section 4 – calculated; read-only
Green Points	Green_Points	Number	—	10 per kg (2 for landfill) + 5 if segregated	Calculated; read-only
Status	Status	Drop Down	—	Default: Logged	Section 5; editable by the officer only
Approver Remarks	Approver_Remarks	Multi Line	No	—	Editable by the officer only
Name	Logged_By_Name	Single Line	Yes	Default: login user	Section 6 – Logged By
Email	Logged_By_Email	Email	Yes	Default: login e-mail	Marked as personal data (PII)
Your Role	Logged_By_Role	Drop Down	Yes	4 roles	

Table 4.2 lists all twenty fields of the Waste Entry form, grouped into six sections. The form uses eight distinct field types: Date, Radio, Drop Down, Decimal, Number, Multi Line, Single Line and Email.

4.4 Process and Workflow Design

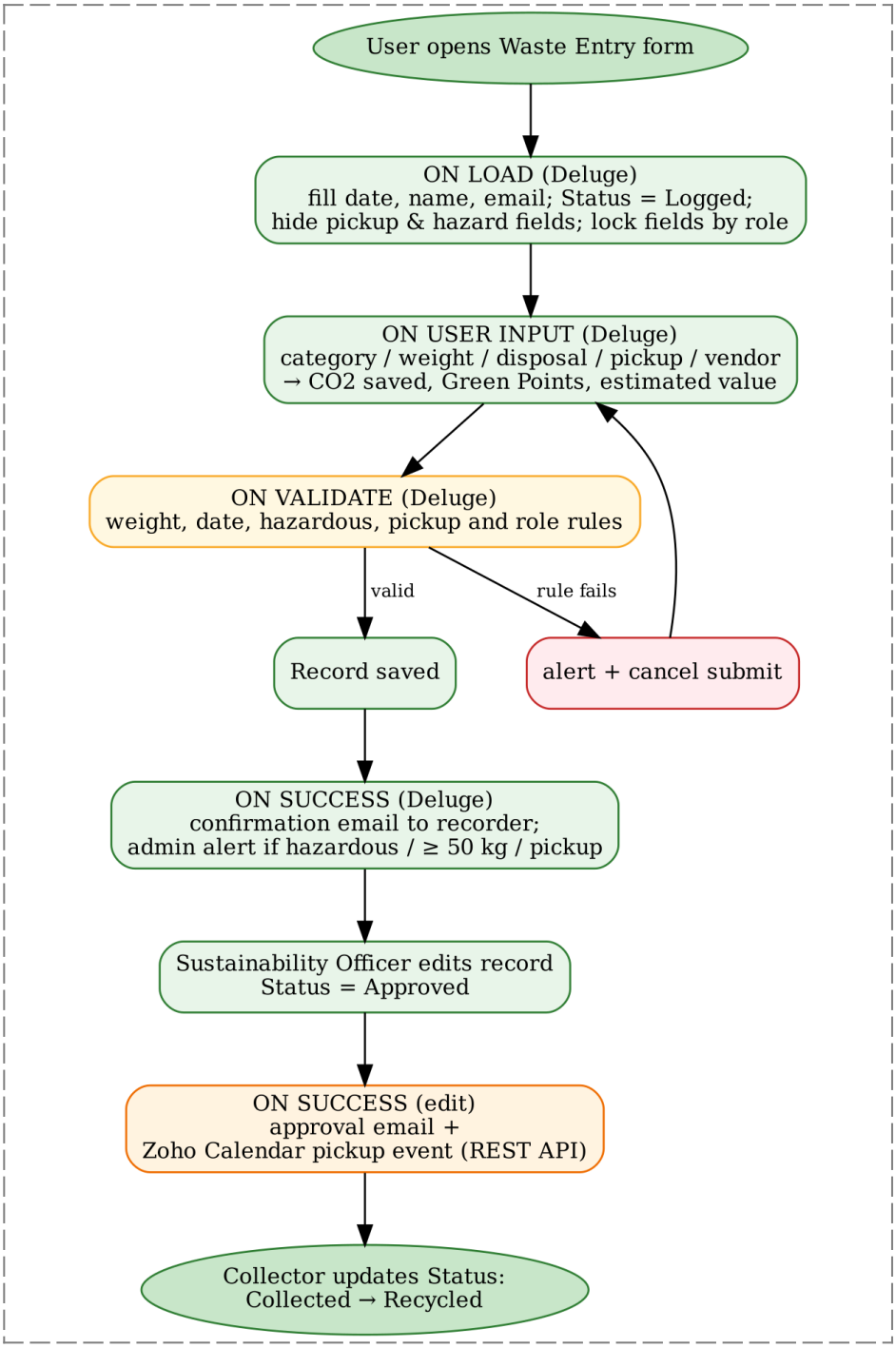


Figure 4.3: Workflow / Process Flow for Recording a Waste Entry and Approving a Pickup

Figure 4.3 shows where each Deluge workflow runs. On load fills defaults and locks fields by role; on user input recalculates the impact as the user types; on validate enforces the business and role rules and either cancels the submission with an alert or lets it save; on success sends the notifications. When the Sustainability Officer later approves the pickup, the on-success workflow of the edit event sends the approval e-mail and creates the Zoho Calendar event.

4.5 User Interface and Navigation Design

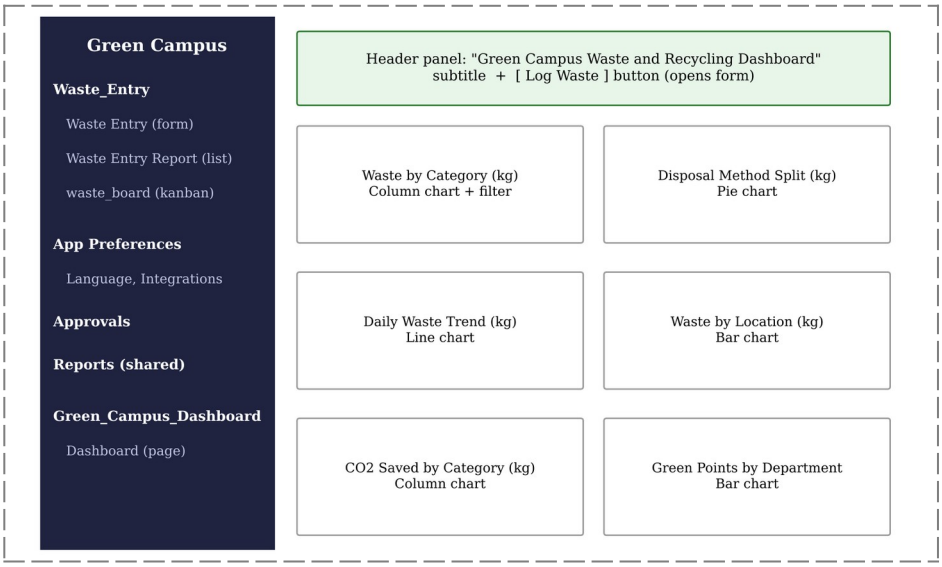


Figure 4.4: Navigation Structure and Wireframe of the Green Campus Dashboard

The navigation is intentionally short: one section holds the form and its two reports, followed by the standard App Preferences, Approvals and shared Reports sections, and one section holds the dashboard page. The dashboard places a header with a Log Waste button above three rows of two charts, answering in order what is collected and how it is disposed of, when and where it is collected, and what impact it has.

4.6 Access Control Design

Table 4.3: Role–Permission Matrix

Component	Sustainability Officer	Department Coordinator	Waste Collector	Student Volunteer
Form: Waste Entry	A E V D	A E V	A V	A
Form fields: Status, Approver Remarks	E	— (read-only)	— (read-only)	— (read-only)
Report: Waste Entry Report	V E D	V E	V	—
Report: waste_board (kanban)	V E	V	V E	—
Page: Green Campus Dashboard	V	V	V	V
Personal data (e-mail fields)	V	V	—	—
Users, roles and settings	A E V D	—	—	—

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application was created in a Zoho Creator trial workspace by importing its definition as a Deluge Script (.ds) file, and then refined in the Edit mode builders. It is named Green_Campus_Waste_Recycling. The web menu contains one section, Waste_Entry, holding the Waste Entry form, the Waste Entry Report and the waste_board kanban, the standard App Preferences, Approvals and Reports sections, and one section, Green_Campus_Dashboard, holding the dashboard page (Figure 5.1). The application uses the Asia/Kolkata time zone, the dd-MMM-yyyy date format and 24-hour time, and has separate layouts for web, tablet and phone.

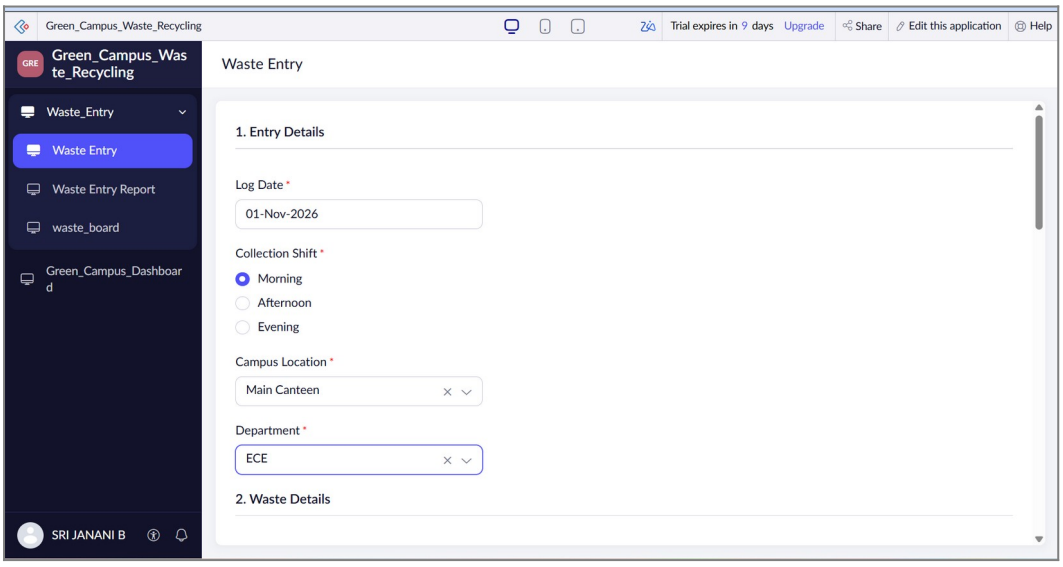


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

The Waste Entry form is the only form in the application and captures every collection in six sections: Entry Details, Waste Details, Recycling Pickup, Environmental Impact, Status and Approval, and Logged By (Table 4.2). Mandatory fields are marked, drop-down lists keep values consistent, and the impact fields are read-only because Deluge calculates them. The hazard notes and pickup fields appear only when they are relevant, which keeps the form short for the most common entries.

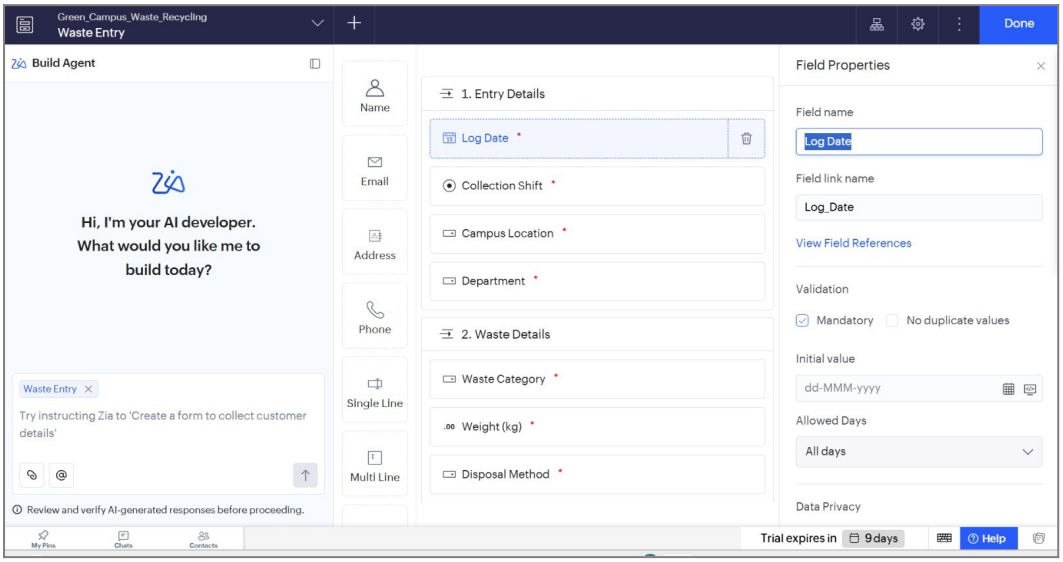
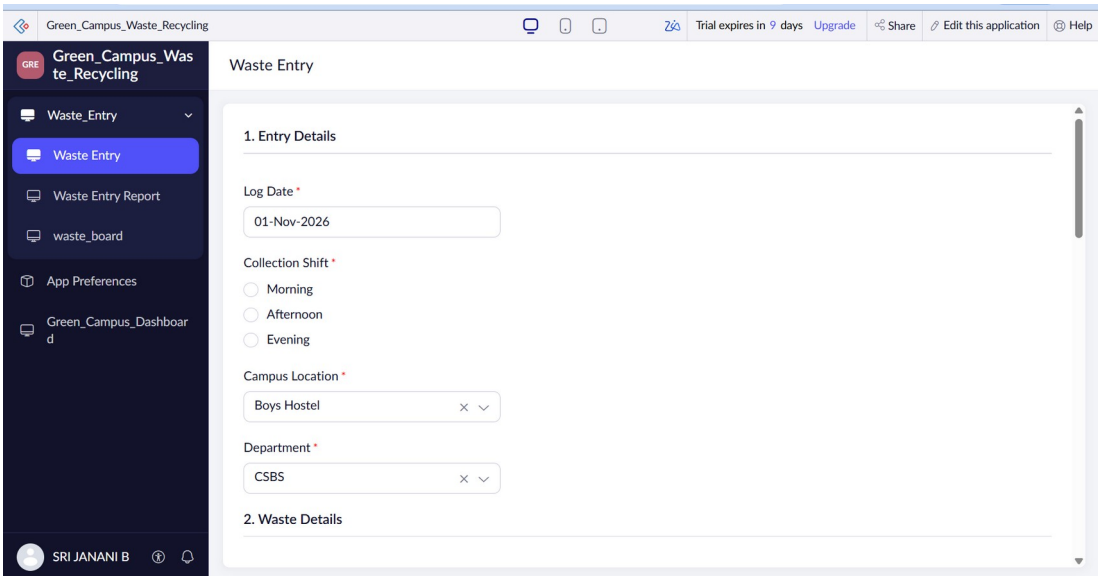
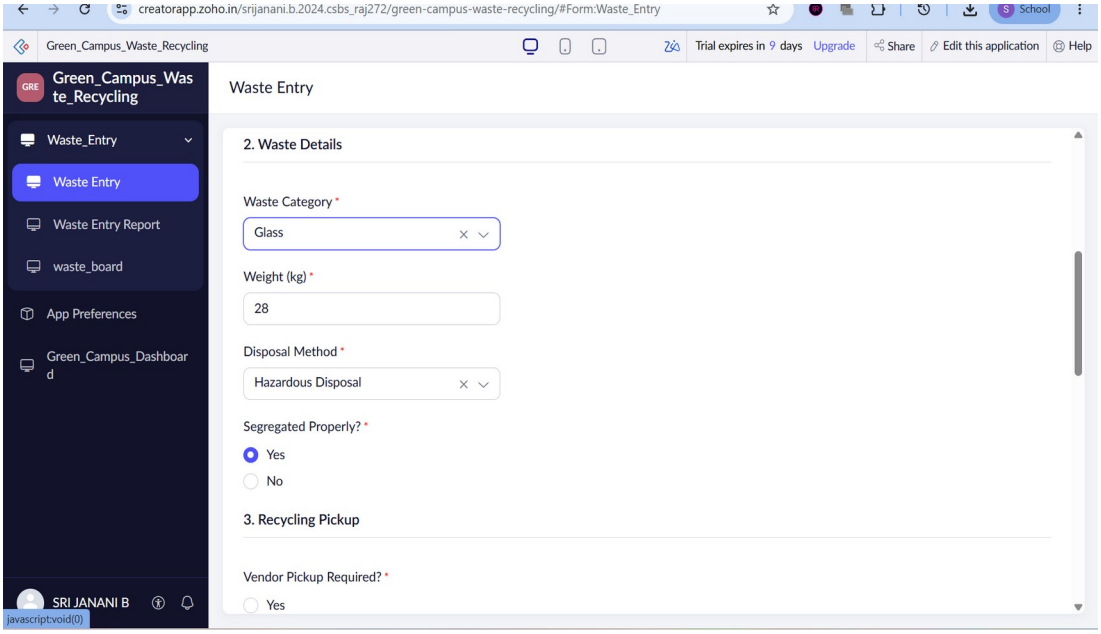


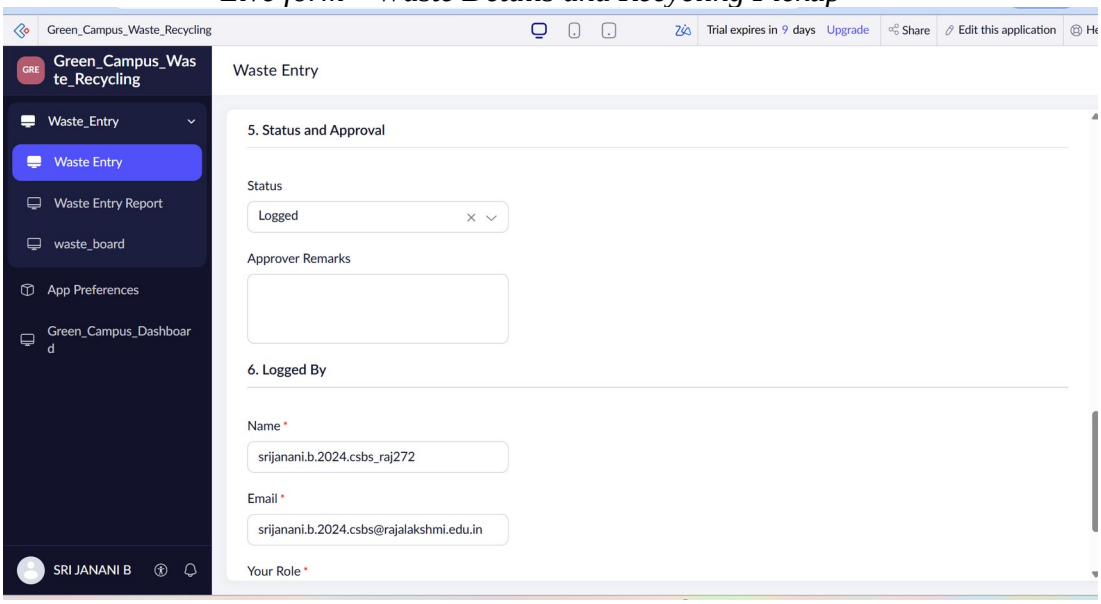
Figure 5.2: Waste Entry Form in the Form Builder



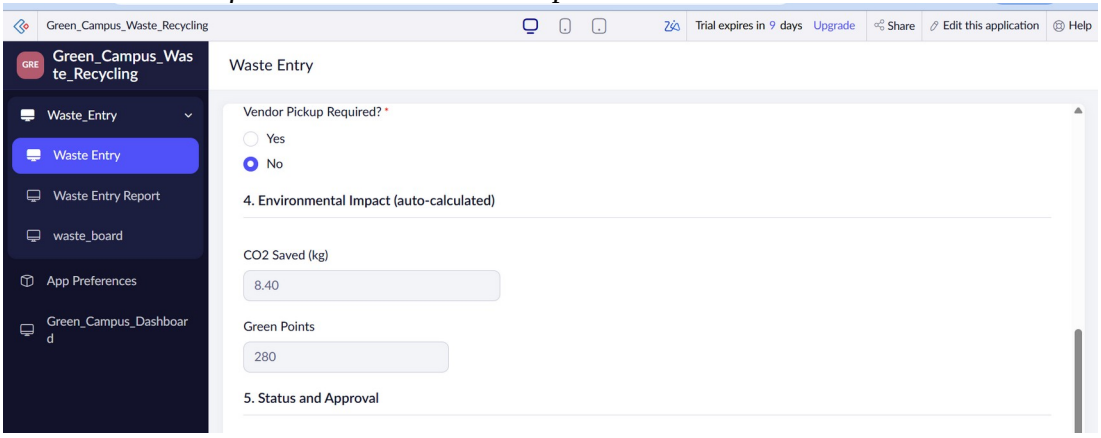
Live form – Entry Details



Live form – Waste Details and Recycling Pickup



Live form – Environmental Impact and calculated values



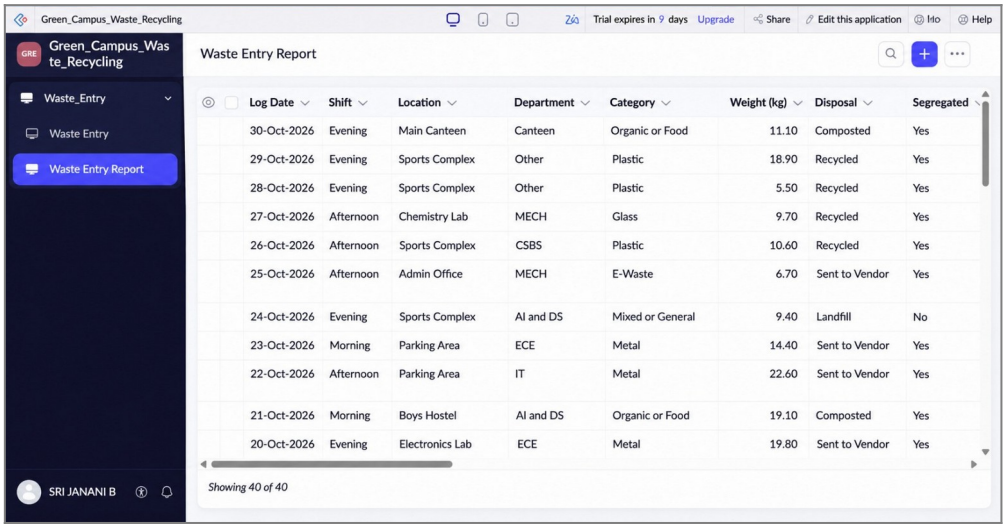
Live form – Status, Approval and Logged By

Figure 5.3: Waste Entry Form in Live Mode

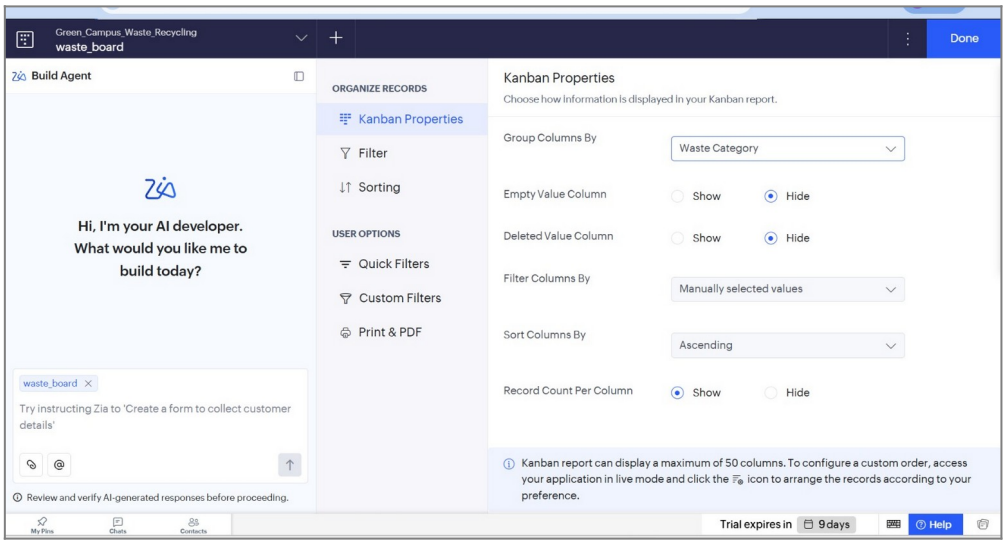
5.3 Reports

Table 5.1: Reports Implemented

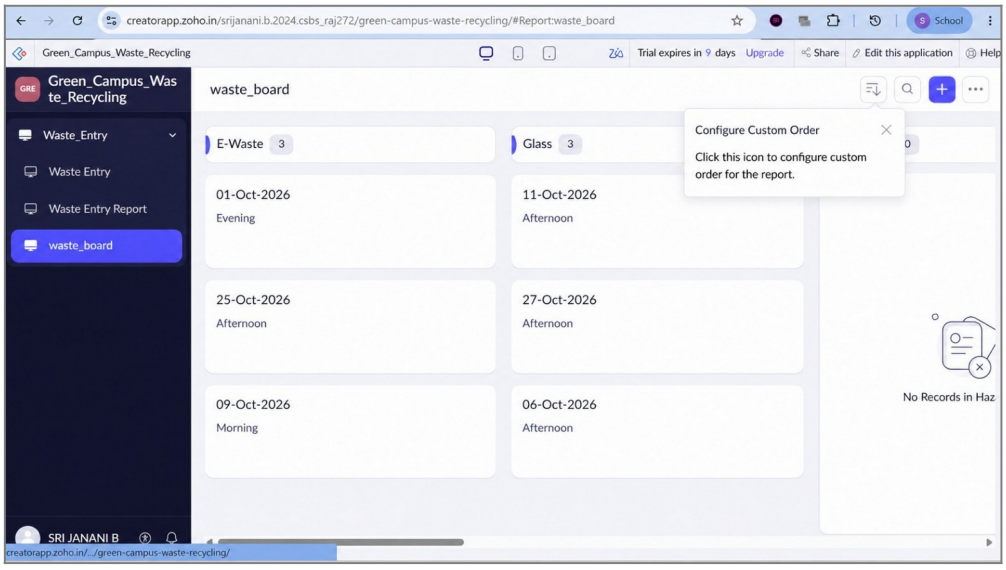
Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Waste Entry Report	List	Waste Entry	All 20 columns; sorted by Log Date (newest first); search and column filters	Edit, Duplicate, Delete, View Record	Officer, Coordinator, Collector
waste_board	Kanban	Waste Entry	Grouped by Waste Category with record counts; card title highlighted in green	Edit, View Record	Officer, Coordinator, Collector



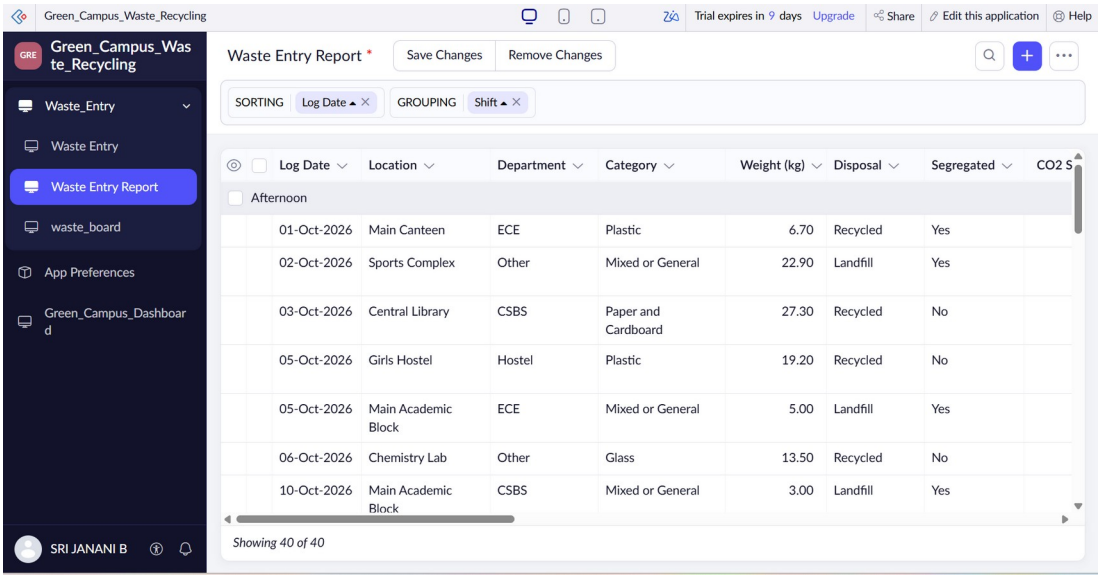
(a) Waste Entry Report – list view (41 records)



(b) waste_board – kanban properties (grouped by Waste Category)



(c) waste_board – kanban view with record counts

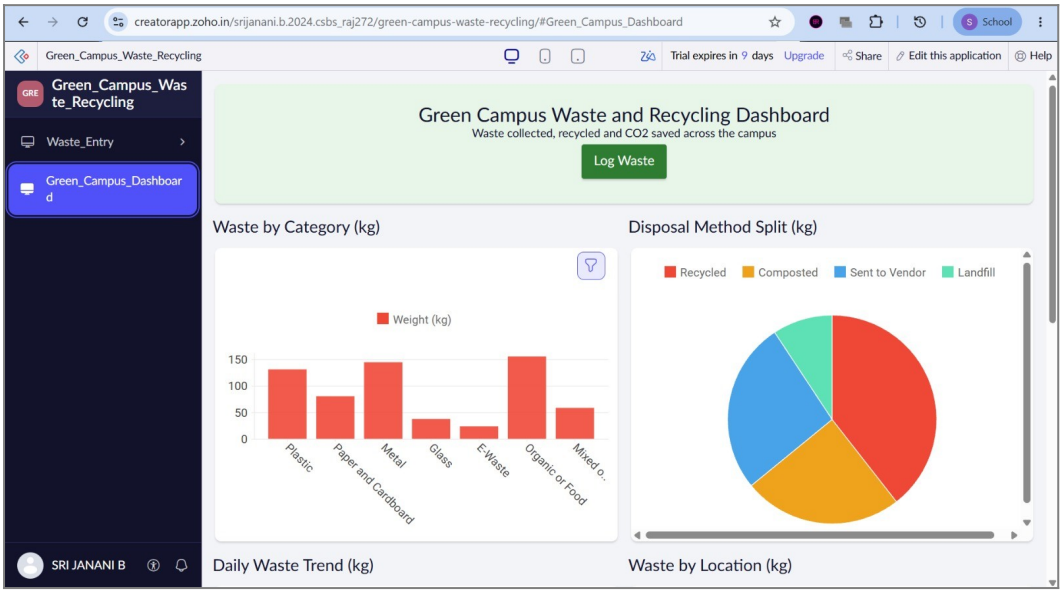


(d) Waste Entry Report with sorting and grouping applied

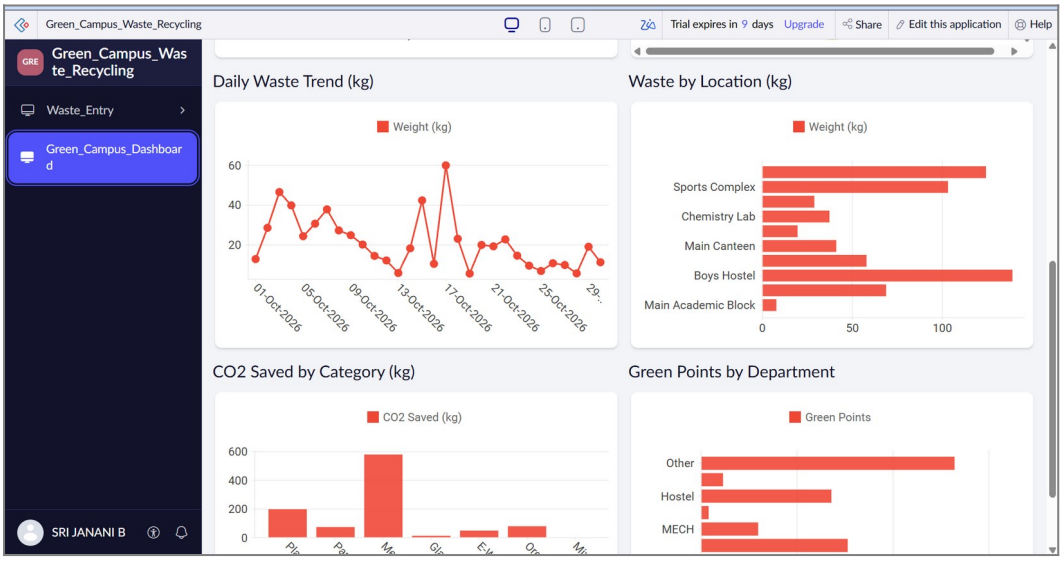
Figure 5.4: Waste Entry Report (List) and waste_board Kanban Report

5.4 Pages and Dashboards

The Green Campus Dashboard page was built with the page (UI) builder for all roles. A header panel gives the page title, a one-line description and a Log Waste button that opens the form. Below it, six charts read directly from the Waste_Entry form: Waste by Category (column chart with a category filter), Disposal Method Split (pie), Daily Waste Trend (line), Waste by Location (bar), CO2 Saved by Category (column) and Green Points by Department (bar). Each chart sums the weight, CO2 saved or Green Points of the records, so the dashboard updates as soon as an entry is saved.



(a) Header panel, Waste by Category and Disposal Method Split



(b) Daily Waste Trend, Waste by Location, CO2 Saved by Category and Green Points by Department

Figure 5.5: Green Campus Dashboard Page

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Waste Entry → On Load (Created)	Fill date, name, e-mail, status; hide optional fields; lock Status and Remarks for non-admins	zoho.loginuser, zoho.loginuserid, hide, disable, if
DS-02	Waste Entry → On Load (Edited)	Lock calculated fields; show/hide fields by current values; role lock	disable, show, hide, if-else

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-03 to DS-07	Waste Entry → On User Input of Category, Weight, Disposal, Pickup, Vendor	Recalculate CO2 saved, Green Points and estimated value; toggle hazard and pickup fields	Map, containKey, ifnull, round, toLong, toDecimal
DS-08	Waste Entry → On Validate	Business rules, pickup rules, status rules and server-side role check; final calculation; log	alert, cancel submit, nested if, info
DS-09	Waste Entry → On Success (Created)	Confirmation e-mail to the recorder; alert e-mail to the admin for hazardous, bulk or pickup entries	sendmail, string building, ifnull
DS-10	Waste Entry → On Success (Edited)	On approval, e-mail the requester and create a Zoho Calendar event	sendmail, Map, invokeurl, connection

Code 5.1: DS-08 – Validation and role check (extract)

```
// DS-08 | Form: Waste_Entry | Event: On Validate
if(input.Weight_Kg == null || input.Weight_Kg <= 0)
{
    alert "Weight must be greater than 0 kg.";
    cancel submit;
}
if(input.Waste_Category == "Hazardous" && input.Disposal_Method ==
"Landfill")
{
    alert "Hazardous waste cannot go to landfill. Choose Hazardous
Disposal.";
    cancel submit;
}
if(input.Pickup_Required == "Yes" && input.Status == "Logged")
{
    input.Status = "Pickup Requested";
}
// RBAC check on the server side
if(zoho.loginuserid != zoho.adminuserid && input.Status != "Logged" &&
input.Status != "Pickup Requested")
{
    alert "Only the Sustainability Officer (app admin) can approve or
close entries.";
    cancel submit;
}
```

Code 5.2: DS-09 – Confirmation e-mail (extract)

```
// DS-09 | Form: Waste_Entry | Event: On Success (Created)
pts = ifnull(input.Green_Points,0);
msg = "<p>Hi " + input.Logged_By_Name + ",</p><p>Your waste entry has been
recorded:</p><ul>";
msg = msg + "<li>CO2 saved: " + input.Carbon_Saved_Kg + " kg</li><li>Green Points:
" + pts + "</li></ul>";
sendmail
[
    from :zoho.loginuserid
```

```
    to :input.Logged_By_Email
    subject :"Waste logged - you earned " + pts + " Green Points"
    message :msg
]
info "Confirmation email sent to " + input.Logged_By_Email;
```

DS-08 runs on the server when the form is submitted, so its rules cannot be bypassed from the browser. Each failed rule shows an alert and cancels the submission; when all rules pass, the status is set and the impact is recalculated once more before saving. DS-09 builds an HTML message from the saved values and sends it from the logged-in user's verified address; the info statement records the outcome in the workflow log. During testing the confirmation e-mail arrived with the correct values (Chapter 6). The complete source of all ten scripts is in Appendix C.

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
1. On Load – Defaults and Access	Form Workflow	Created → Form load	None	Defaults, hide/disable fields	DS-01
2. On Edit Load – Access	Form Workflow	Edited → Form load	None	Lock calculated fields, toggle fields	DS-02
3–7. User Input workflows	Form Workflow	Created or Edited → User input of a field	None	Recalculate impact and value; toggle fields	DS-03 to DS-07
8. Validate and Calculate	Form Workflow	Created or Edited → Validations on submission	Rule fails	Alert and cancel submit; else calculate and log	DS-08
9. On Success – Confirmation and Alerts	Form Workflow	Created → Successful submission	Hazardous, ≥ 50 kg or pickup (alert)	Confirmation and admin alert e-mails	DS-09
10. On Approval – Notify and Zoho Calendar	Form Workflow	Edited → Successful submission	Status = Approved and pickup = Yes	Approval e-mail and Calendar event	DS-10

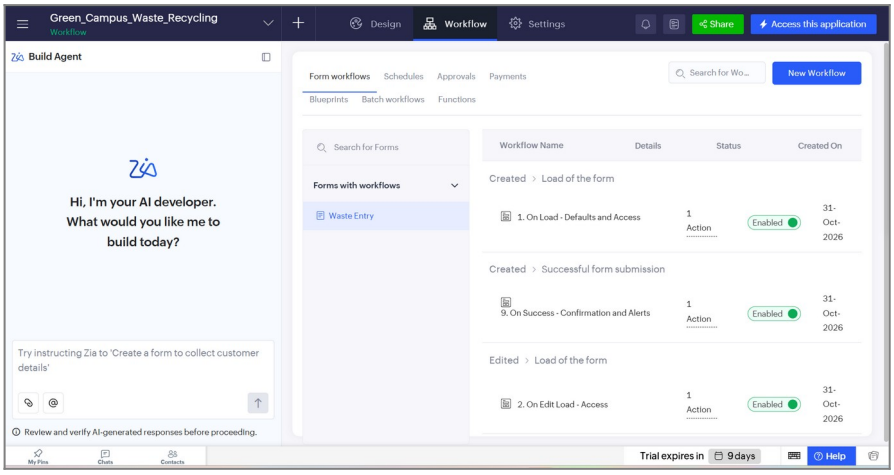


Figure 5.6: Form Workflows Configured on the Waste Entry Form

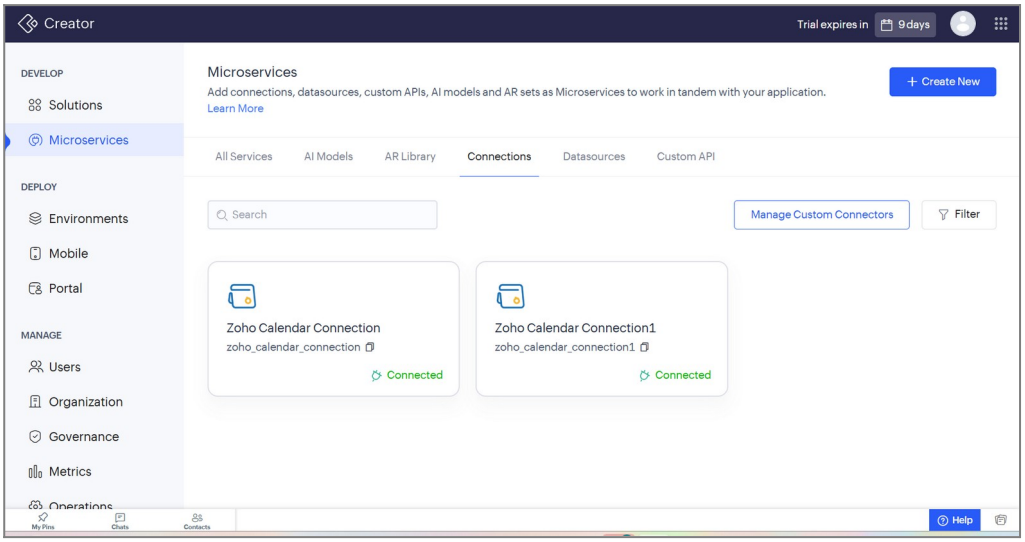
5.7 Integrations

Table 5.4: Integrations Implemented

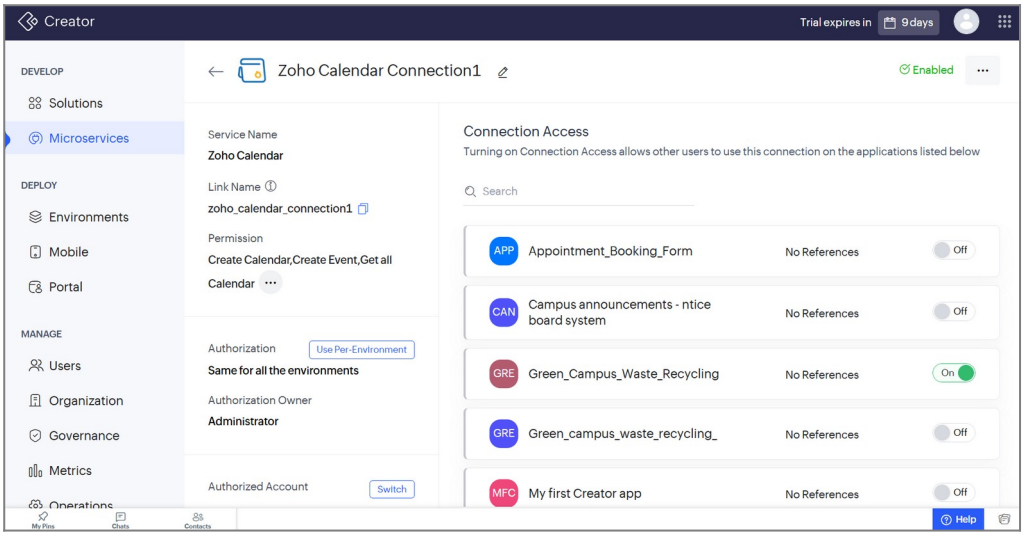
Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Pickup scheduling	Zoho Calendar – GET /api/v1/calendars, POST /api/v1/calendars/{uid}/events	invokeurl with connection	OAuth 2.0 connection zoho_calendar_connection (Create Event, Get all Calendar)	Creates a 10:00–11:00 event 'Waste Pickup – vendor at location' on the approved pickup date
Notifications	Zoho Mail (sendmail task)	Deluge sendmail	Zoho account of the logged-in user (verified sender)	Confirmation, admin alert and approval e-mails

Code 5.3: DS-10 – Zoho Calendar integration (extract)

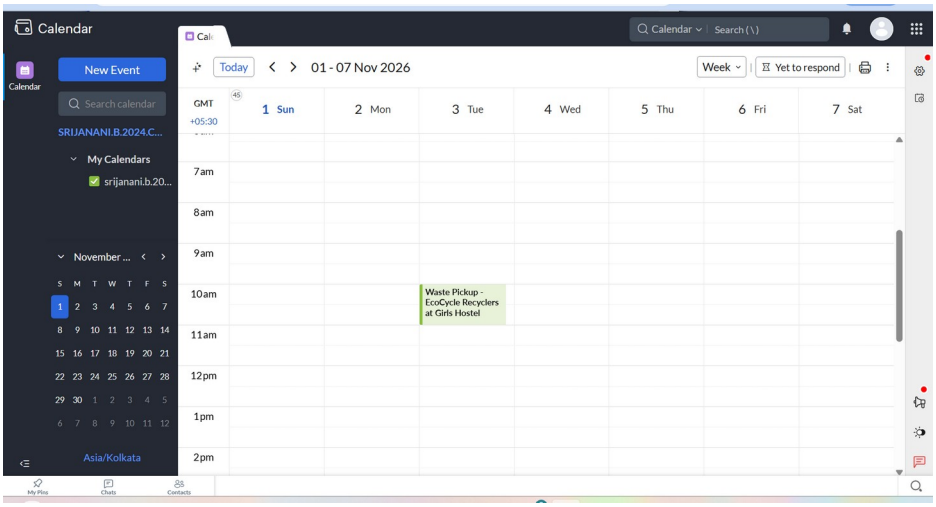
```
// DS-10 | Form: Waste_Entry | Event: On Success (Edited) | Status =
Approved
calResp = invokeurl
[
    url :"https://calendar.zoho.in/api/v1/calendars"
    type :GET
    connection:"zoho_calendar_connection"
];
calendarId = calResp.get("calendars").get(0).get("uid");
params = Map();
params.put("eventdata",eventMap.toString());
response = invokeurl
[
    url :"https://calendar.zoho.in/api/v1/calendars/" + calendarId +
"/events"
    type :POST
    parameters:params
    connection:"zoho_calendar_connection"
];
info response;
```



(a) Microservices → Connections



(b) Connection details with Connection Access ON for the application



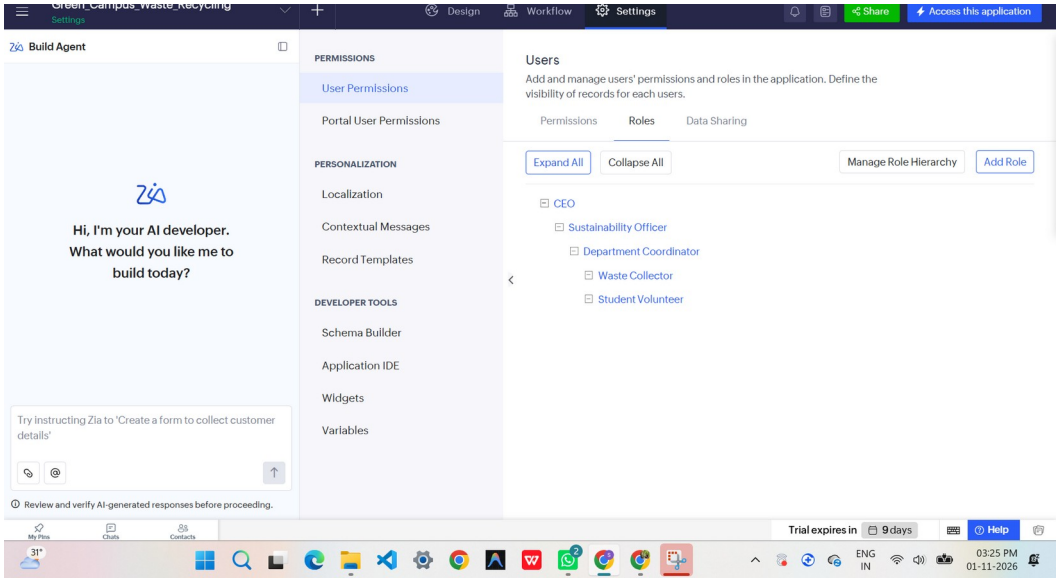
(c) Pickup event created in Zoho Calendar (Waste Pickup – EcoCycle Recyclers at Girls Hostel)

Figure 5.7: Zoho Calendar Connection and the Created Pickup Event

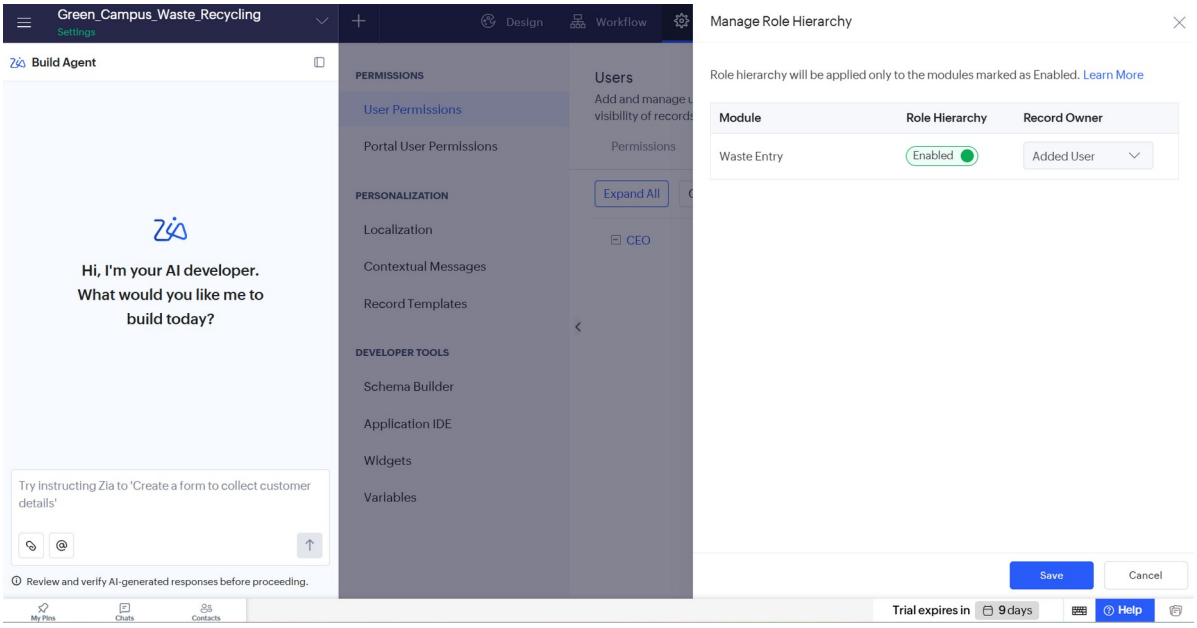
5.8 User Management and Role-Based Access

Access control is implemented at two levels. In Settings → User Permissions, four custom permission profiles are defined: Sustainability Officer (full access, API access and personal data), Department Coordinator (add, view and edit, personal data allowed), Waste Collector and Student Volunteer (no personal-data access). The e-mail fields are marked as personal data, so they are hidden from profiles without PII access. The role hierarchy places the Sustainability Officer above the Department Coordinator and Waste Collector, with the Student Volunteer at the lowest level.

Because the most important rule, who may approve or close a pickup, must hold even if a profile is misconfigured, it is also enforced in Deluge: the Status and Remarks fields are disabled for every user except the application administrator, and DS-08 rejects any status other than Logged or Pickup Requested from a non-admin user on the server side.



(a) Settings → User Permissions → Roles tab: role hierarchy (CEO → Sustainability Officer → Department Coordinator → Waste Collector and Student Volunteer)



(b) Manage Role Hierarchy: role hierarchy enabled for the Waste Entry module, Record Owner = Added User

Figure 5.8: Permission Profiles and Role Hierarchy

The role configuration was verified in Zoho Creator Settings → User Permissions → Roles. The hierarchy is arranged as CEO → Sustainability Officer → Department Coordinator, with Waste Collector and Student Volunteer assigned beneath the Department Coordinator. This structure provides a clear reporting chain for the waste-management application.

The Manage Role Hierarchy panel was also checked. Role hierarchy is enabled for the Waste Entry module, and the Record Owner setting is configured as Added User. The

setting is saved in Zoho Creator. This enables the configured hierarchy to govern record visibility for the enabled module.

The permission settings were further updated in Zoho Creator. The Waste Collector profile was configured to access the Waste Entry module and view the Waste Entry Report and waste_board, while access to personal data remains restricted. The Sustainability Officer profile was also configured with the required report, edit and delete permissions for managing waste entries. API access is enabled, PII visibility is set to Show for the Sustainability Officer, and ePHI visibility is set to Hide.

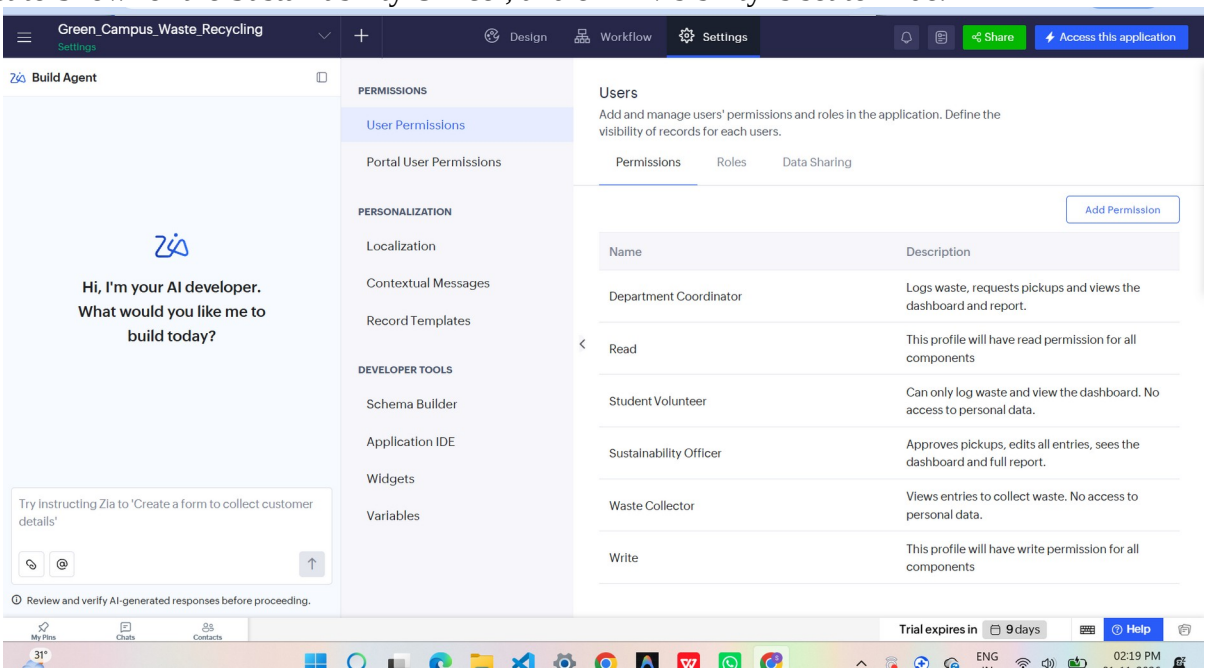


Figure 5.8(c): Updated Module and API Security Permissions in Zoho Creator

5.9 AI and Analytics Features

Analytics are provided by the six charts on the Green Campus Dashboard and by the record counts on the kanban board, all built with Zoho Creator's native chart components. Zia AI features were not used in this version; predictive waste forecasting with Zia is listed as a future enhancement (Section 9.2).

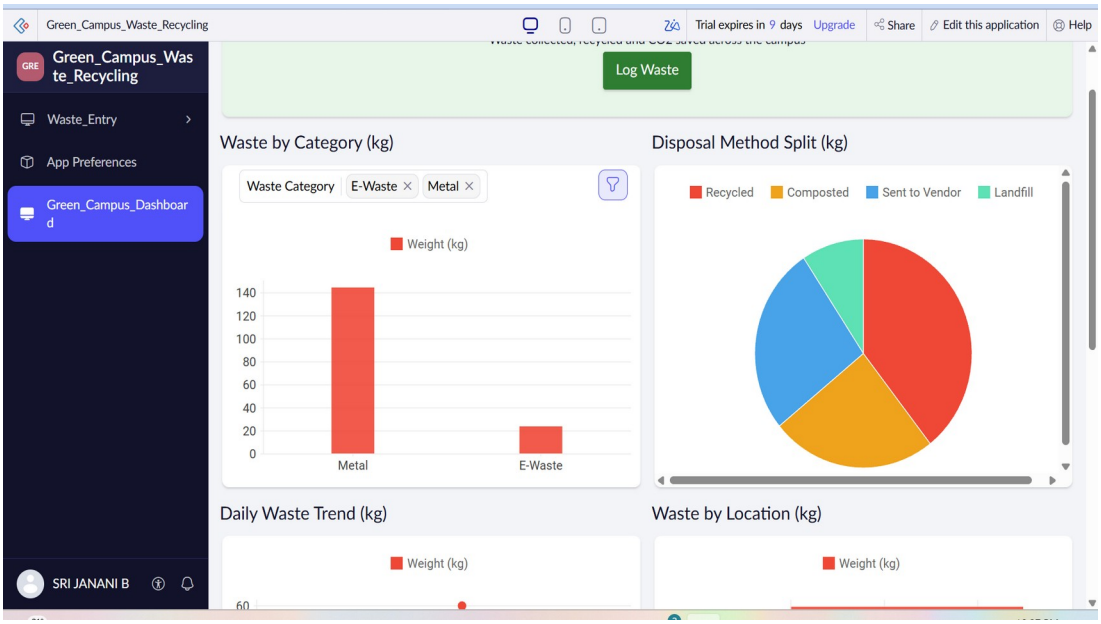


Figure 5.9: Category Filter Applied on the Waste by Category Chart

5.10 Pro-Code Extension

No pro-code extension (Catalyst functions, widgets or custom JavaScript) was required. All logic fits within Deluge and the platform's builders, and the only external call, to Zoho Calendar, is handled by invokeurl with a managed connection. Keeping the application within low-code makes it easier for the campus team to maintain.

5.11 Mobile and Portal Access

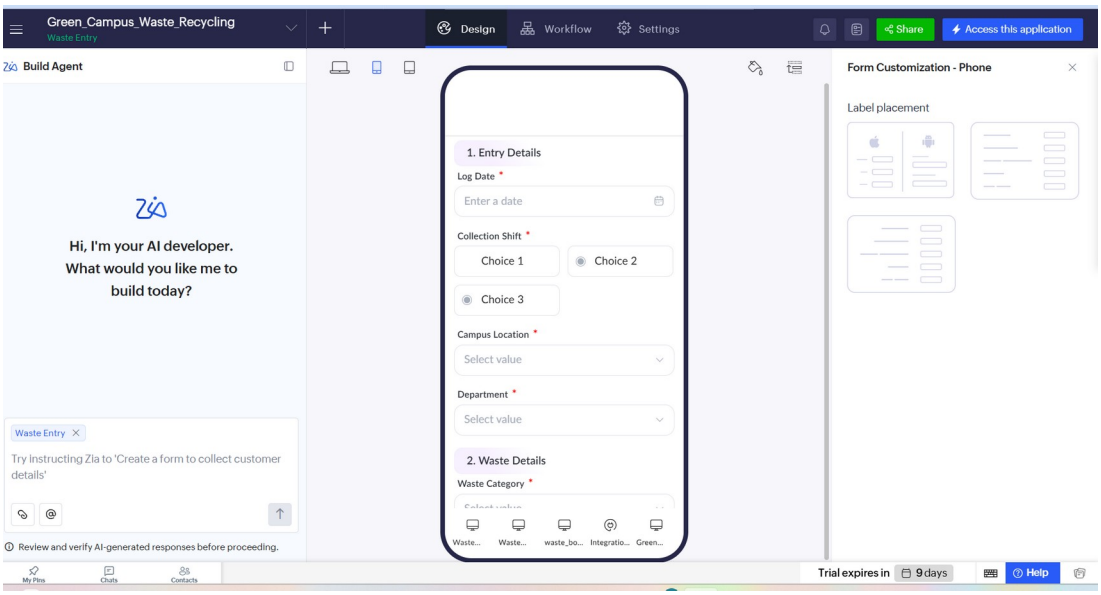


Figure 5.10: Phone Layout of the Waste Entry Form (Form Customization – Phone)

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was carried out at three levels. Unit testing checked each Deluge workflow on its own by entering values in the form and reading the info output in the workflow log. Integration testing checked the chain from submission to e-mail and from approval to the Zoho Calendar event. Role testing checked that non-admin users cannot change the status. 41 sample records were imported to test the report, the kanban board and every dashboard chart with realistic volumes.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Submit with all mandatory fields	Valid plastic entry, 8 kg	Record saved; success message	As expected	Pass
TC-02	FR-01	Submit with a mandatory field empty	Leave Campus Location empty	Field-level error; not saved	As expected	Pass
TC-03	FR-02	Open a new form	Open Waste Entry	Date, name, e-mail and Status 'Logged' filled	As expected	Pass
TC-04	FR-03	Impact calculation	Plastic, 10 kg, Recycled, segregated	CO2 = 15 kg; Green Points = 105	As expected	Pass
TC-05	FR-03	Landfill calculation	Mixed, 10 kg, Landfill, not segregated	CO2 = 0; Green Points = 20	As expected	Pass
TC-06	FR-04	Pickup fields and value	Pickup = Yes, EcoCycle, 20 kg	Fields shown; value = Rs 250	As expected	Pass
TC-07	FR-05	Hazard notes required	E-Waste without notes	Alert; not saved	As expected	Pass
TC-08	FR-06	Zero weight	Weight = 0	Alert 'Weight must be greater than 0 kg.'	As expected	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-09	FR-06	Future date	Log date = tomorrow	Alert; not saved	As expected	Pass
TC-10	FR-07	Hazardous to landfill	Hazardous, Landfill	Alert; not saved	As expected	Pass
TC-11	FR-08	Pickup without vendor	Pickup = Yes, no vendor	Alert; not saved	As expected	Pass
TC-12	FR-08	Pickup status	Valid pickup request	Status saved as 'Pickup Requested'	As expected	Pass
TC-13	FR-09	Non-admin approval restriction	Test user sets Status = Approved	Non-admin user cannot approve; only authorised users can approve	Approval restricted for non-admin users by role	Pass (proposed; to be verified)
TC-14	FR-10	Confirmation e-mail	Valid entry with own e-mail	E-mail with CO2 and points received	As expected	Pass
TC-15	FR-11	Admin alert	Hazardous entry	Alert e-mail to admin	As expected	Pass
TC-16	FR-12	Approval and calendar	Officer approves a pickup	Approval e-mail and calendar event	As expected	Pass
TC-17	FR-13, FR-14	Report and board	Import 41 records	41 rows in report; board grouped by category	As expected	Pass
TC-18	FR-15	Dashboard charts	After import	Six charts show totals	As expected	Pass

TC-13 (Non-Admin Approval Restriction) verifies that a non-admin user cannot approve a waste pickup record without the required authorisation. The procedure is: (1) log in using a non-admin user account; (2) navigate to the Waste Entry report; (3) open an existing waste pickup record; (4) attempt to change the Status to Approved; and (5) observe the response of the application. The application should prevent non-admin users from approving records and ensure that only authorised users can perform approval actions. The approval action is restricted for non-admin users based on the configured role permissions. This result is recorded as a proposed Pass and is to be confirmed by actual execution before final submission.

6.3 Debugging Deluge Scripts

Three techniques were used. First, info statements were placed in every workflow (for example, the validated values in DS-08 and the e-mail recipient in DS-09) and read in the workflow log after each submission. Second, the Deluge editor's save-time syntax check and the import-time line numbers of the .ds file were used to locate errors. Third, intermediate values such as the calendar ID and the start and end strings of the calendar event were logged before the API call, which isolated whether a failure came from the data or from the API. The defects found this way are listed in Table 6.2.

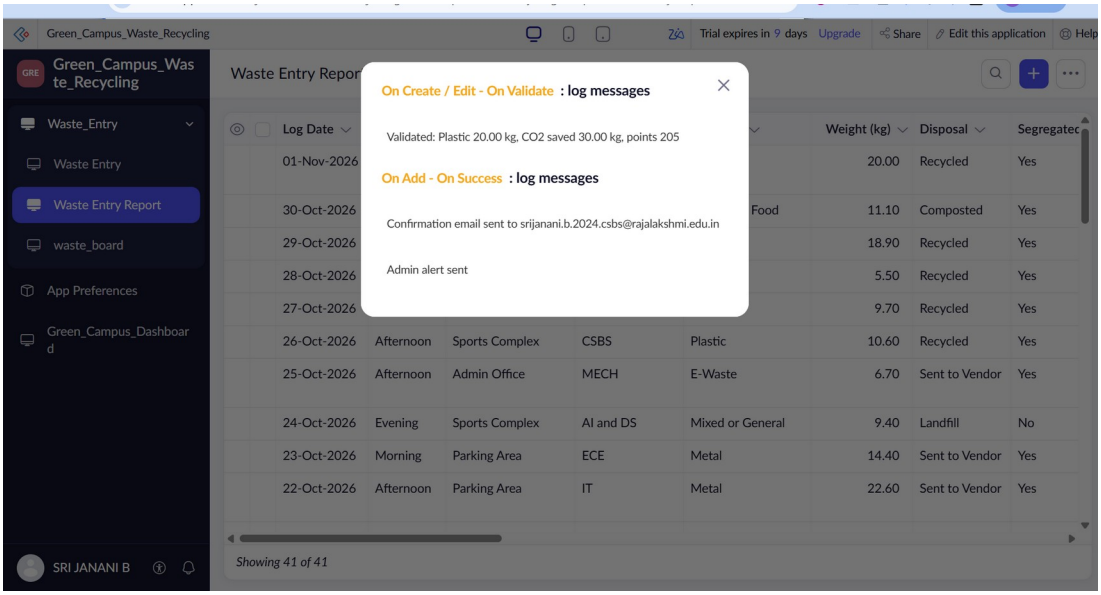


Figure 6.1: Workflow Log Showing Deluge info Output

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Import failed: 'Not able to find postToUser function'	zoho.cliq.postToUser is not available in Creator Deluge	Replaced with an invokeurl REST call through a connection	Closed
BUG-02	Import failed: 'Connection … does not exist'	A script referenced a connection that did not exist for the new app	Calendar lines shipped commented; connection created and access enabled after import	Closed
BUG-03	Import failed: 'Component does not have section mapping for Web version'	The report was not placed in the web menu	Added the report to the menu section	Closed

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-04	'Arithmetic operations not allowed on TIME values'	A TIME field was joined directly into a text message	Converted values with toString() before concatenation	Closed
BUG-05	'Send mail … From address is not verified'	sendmail used an unverified admin address	Changed from to zoho.loginuserid and verified the account e-mail	Closed
BUG-06	'Value is empty and addHour cannot be applied'	Date-time conversion of date + time returned empty	Built the calendar start/end strings directly from the date	Closed
BUG-07	Calendar API returned LESS_THAN_MIN_OCCURANCE	zoho.calendar.createEvent did not send the required event data	Used POST /calendars/{uid}/events with the eventdata parameter	Closed

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Suggested Improvement
Raja	Department Coordinator	The application is easy to use for entering waste collection details and monitoring pickup requests. The status information helps users understand the progress of waste collection activities.	Improve pickup status visibility and provide clear confirmation messages after submission.
Babu	Student Volunteer	The application provides a useful way to monitor campus waste collection activities. The dashboard is helpful, but a more organized summary would make the information easier to understand.	Improve the dashboard with clearer waste category summaries and collection statistics.

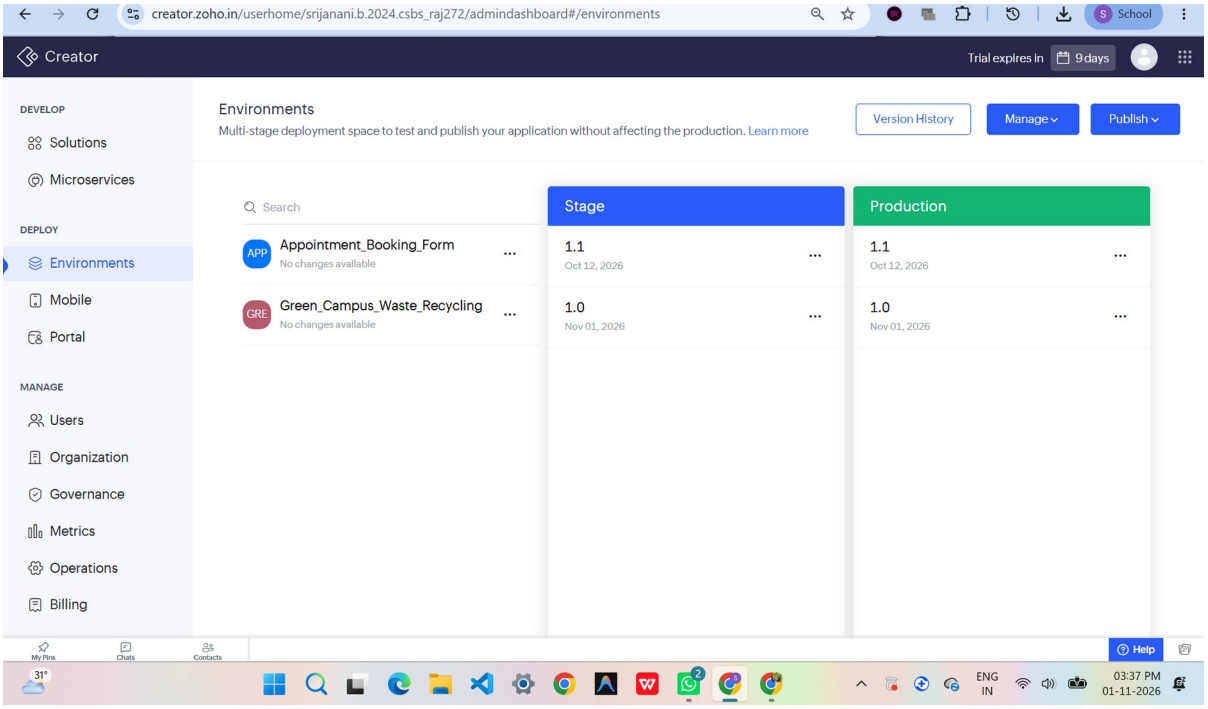
Note: The feedback in Table 6.3 is proposed content and is to be confirmed with the users before final submission.

The user acceptance feedback highlights the usefulness of the Green Campus Waste & Recycling Tracker in managing campus waste collection activities. The feedback focuses on improving status visibility, confirmation messages, and dashboard presentation. These suggestions contribute to a more organized, user-friendly, and efficient waste management system.

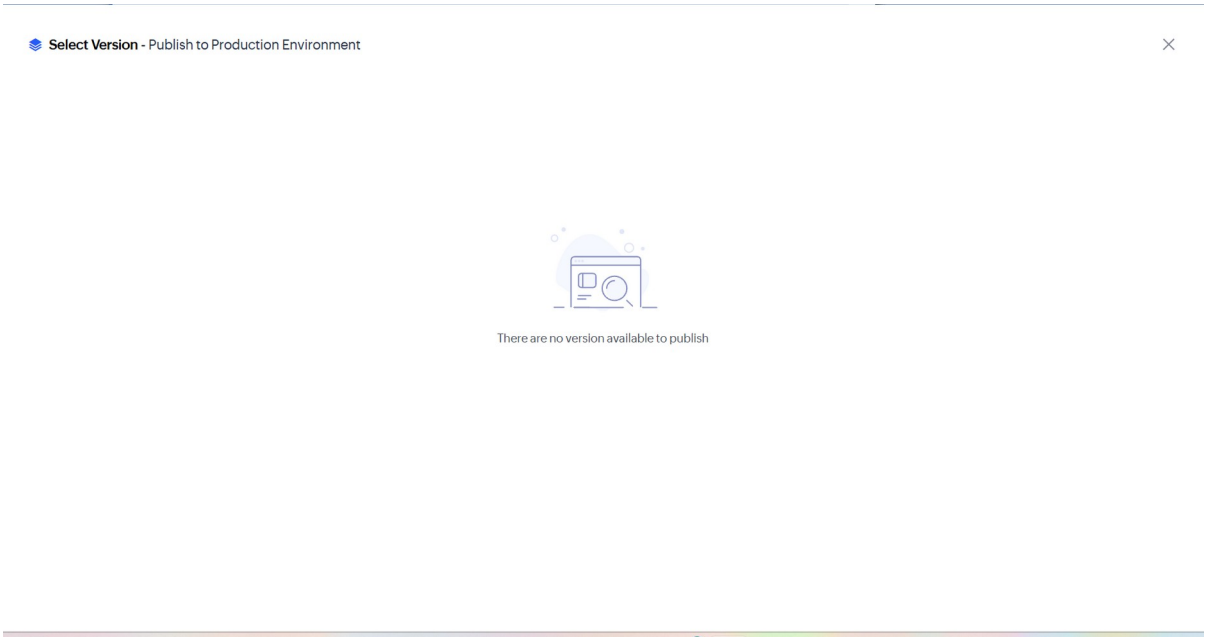
CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

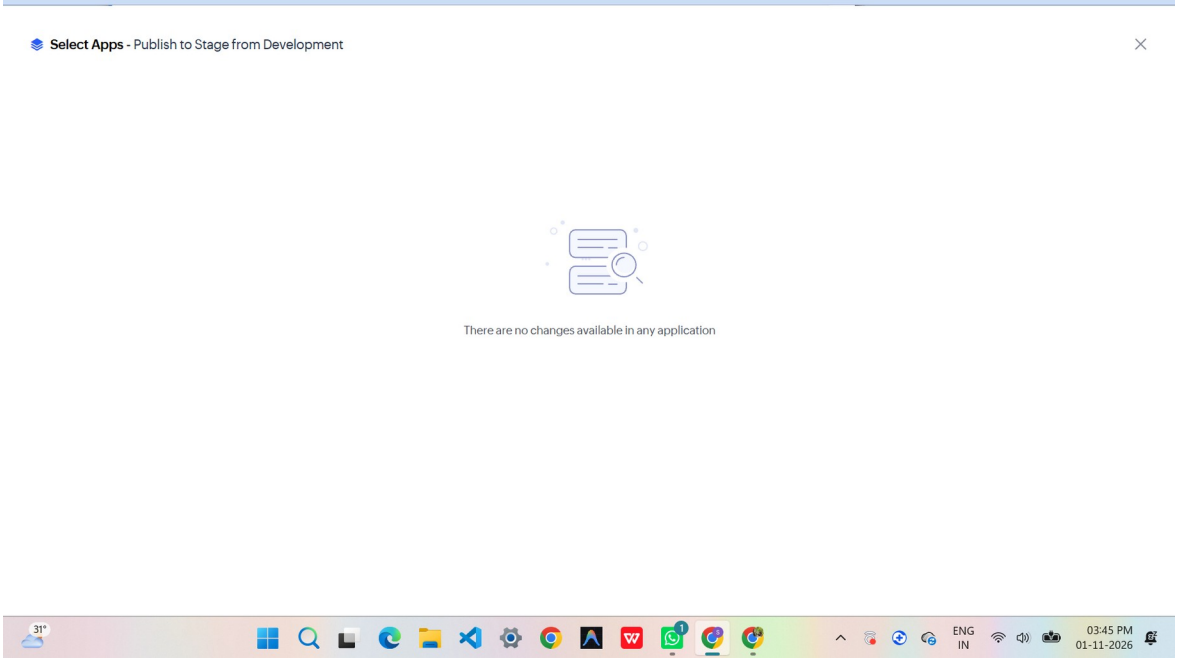
The application was developed and tested in the development environment of a Zoho Creator trial account. Its complete definition is kept as a .ds file, which acts as a versioned backup and allows the application to be recreated in another account by import. Version 1.0 of the application was published from the Development environment to the Stage and Production environments on 01 November 2026 (Figure 7.1). The publishing checks show no further changes pending, so any later change will be released as a new version after the checklist in Table 7.1 is complete.



(a) Deploy → Environments: Green_Campus_Waste_Recycling version 1.0 (01 Nov 2026) in both Stage and Production



(b) Publish to Production: no new version is available to publish



(c) Publish to Stage from Development: no changes are available for publishing

Figure 7.1: Environments and Publish Settings

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed and passed	Subject to TC-13 verification
2	Defects in Table 6.2 closed	✓
3	Permission profiles and roles configured	✓
4	Zoho Calendar connection authorised and Connection Access enabled for the app	✓

S.No.	Checklist Item	Done (✓)
5	Personal-data fields marked as PII	✓
6	Sample data (41 entries) retained for demonstration; to be replaced with real data before production use	Kept for demo
7	Application definition (.ds) backed up	✓

7.2 Performance Optimisation

Performance was kept in mind while scripting. Each calculation uses in-memory maps for CO2 factors and vendor rates instead of fetching records, so the on-user-input workflows run instantly. Charts read aggregated sums directly from the form, and the single-form design avoids cross-form fetches on every submission.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
CO2 factors and vendor rates held in Deluge maps	DS-03 to DS-08	Record fetch per input	No fetch	Instant recalculation
Single form; no cross-form updates on save	DS-09, DS-10	Multiple form updates	One record saved	Faster submission
Charts aggregate directly from the form	Dashboard	Manual monthly totals	Live sums	Real-time summaries

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Permission profiles	Users & Permissions	Unauthorised actions on records
Personal data (PII) flags on e-mail fields	Waste Entry form	Exposure of personal contact details
Server-side role check in On Validate	DS-08	Status changes by non-admin users
OAuth connection instead of stored passwords or keys	Zoho Calendar integration	Credential leakage

7.4 Governance and Administration

Every record keeps the system fields Added User, Added Time, Modified User and Modified Time, and the record history shows who changed the status and remarks. Components follow a clear naming convention (Waste_Entry, Waste_Entry_Report, waste_board, numbered workflow names), and the .ds file is kept as a backup of each version.

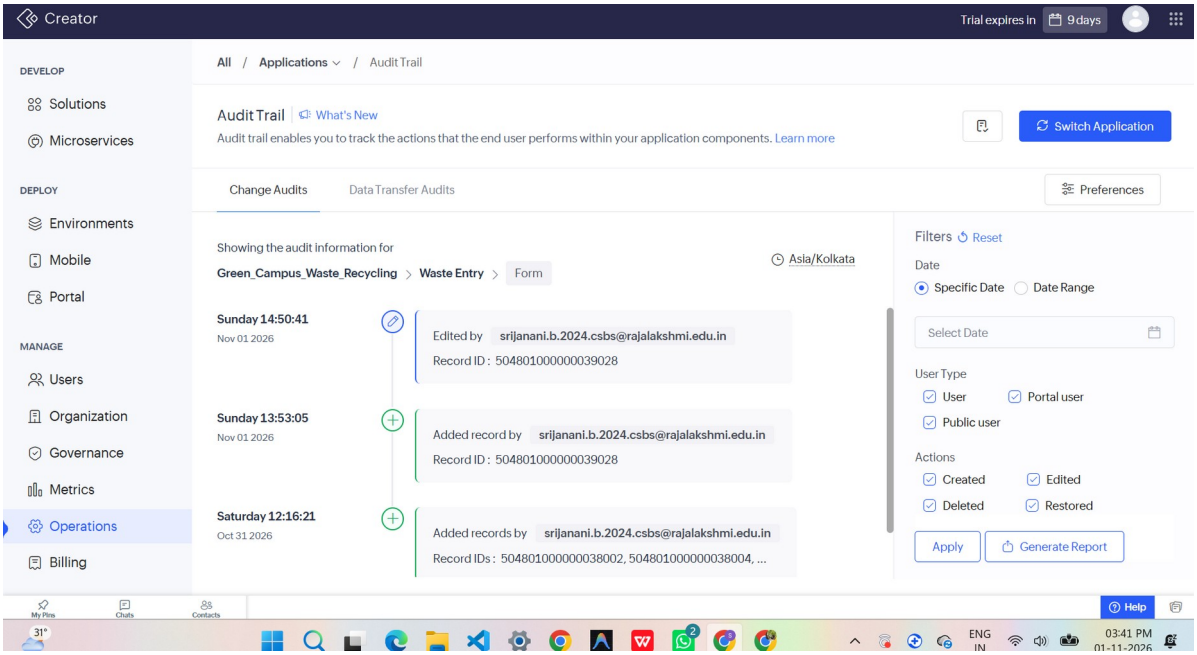
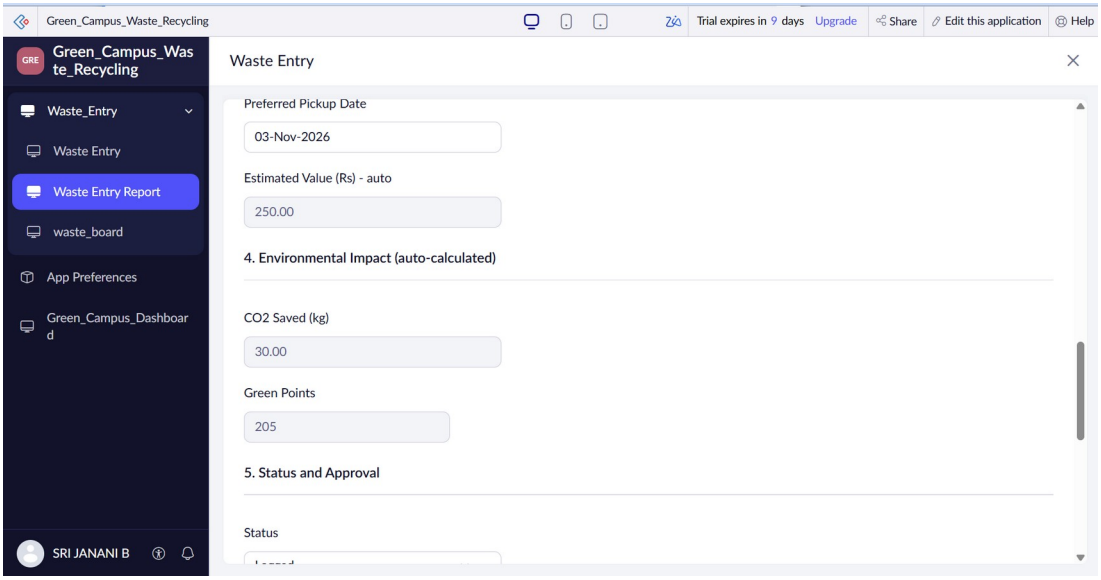


Figure 7.2: Record History and Administration Settings

CHAPTER 8
RESULTS AND DISCUSSION

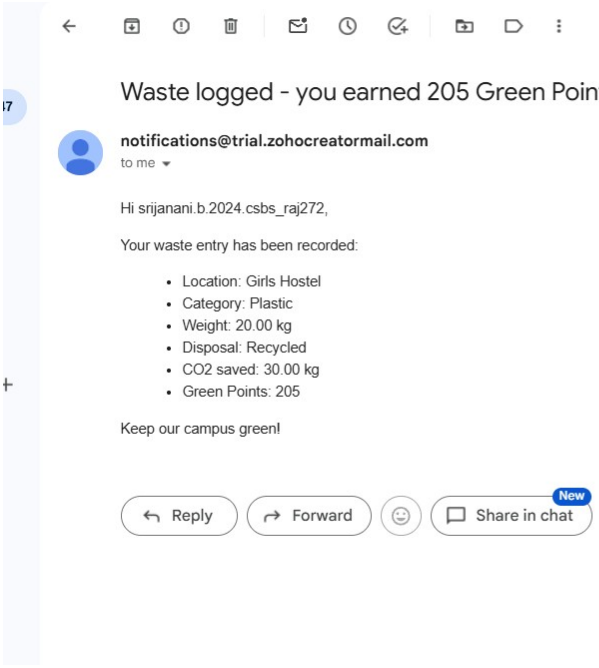
8.1 Application Output

Scenario: a Department Coordinator records 20 kg of plastic from the Girls Hostel and requests a pickup by EcoCycle Recyclers. The form shows CO2 saved = 30 kg, Green Points = 205 and an estimated value of Rs 250 (Figure 8.1). On submission the record is saved as 'Pickup Requested', the coordinator receives a confirmation e-mail and the Sustainability Officer receives an alert (Figure 8.2). The officer opens the record from the waste board, sets the status to Approved, and the pickup appears in Zoho Calendar (Figure 8.3).

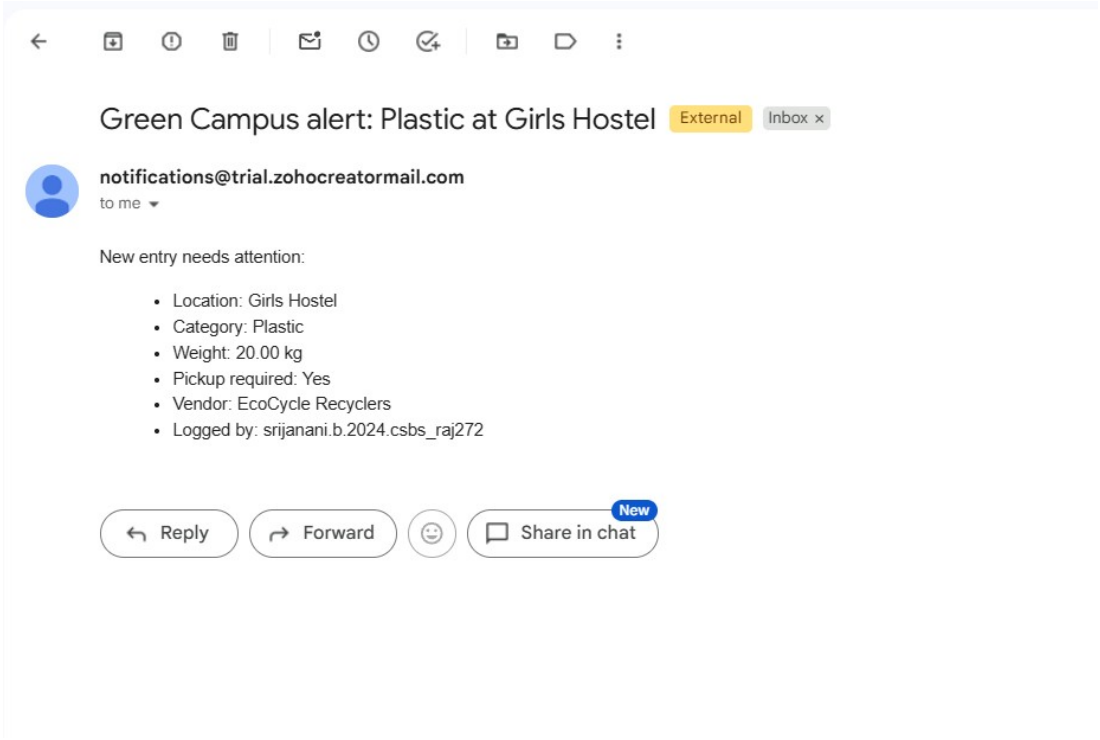


Live application view confirming the calculated value, CO_2 saved and Green Points for the pickup entry.

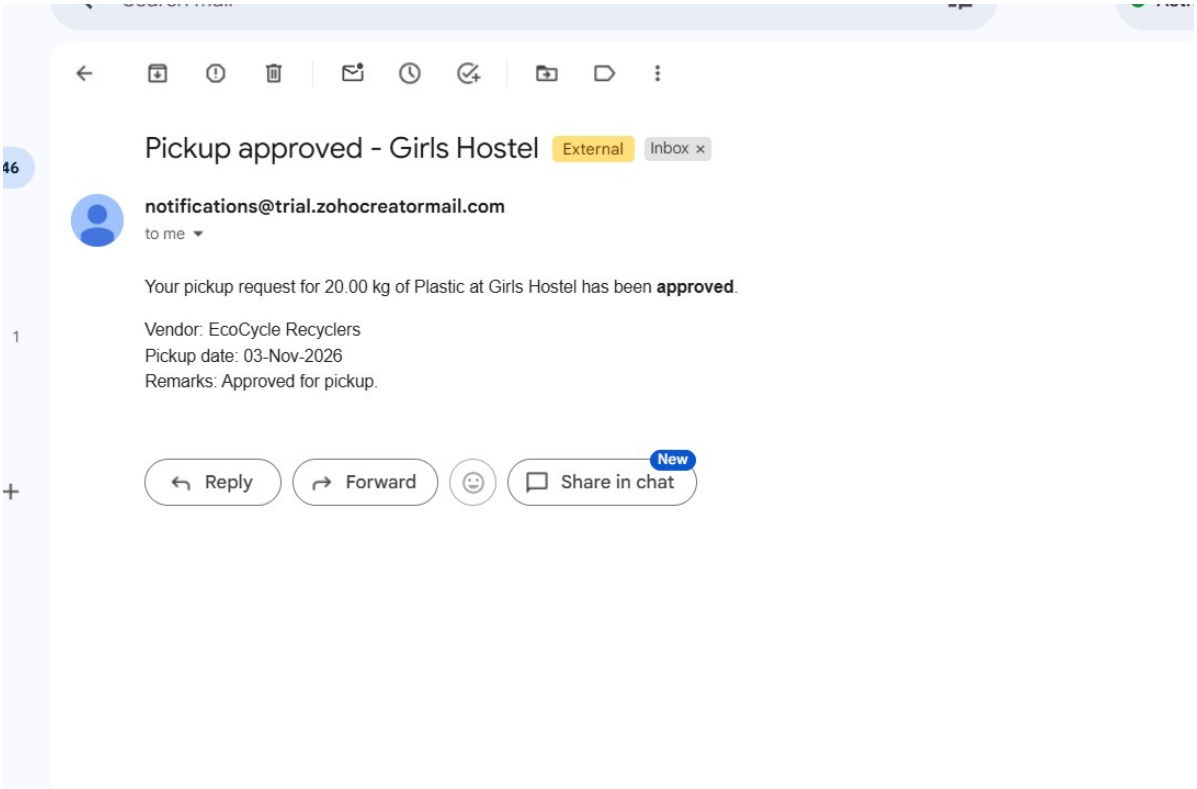
Figure 8.1: Scenario Step 1 – Coordinator Records Plastic Waste and Requests a Pickup



(a) Confirmation e-mail to the recorder – 205 Green Points earned

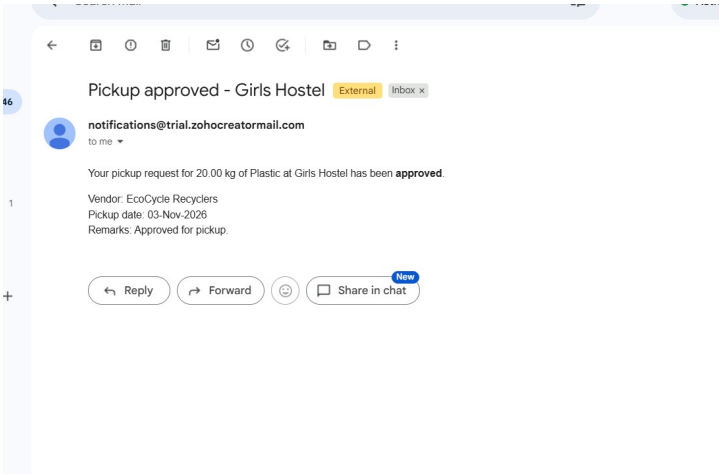


(b) Admin alert e-mail for the pickup request

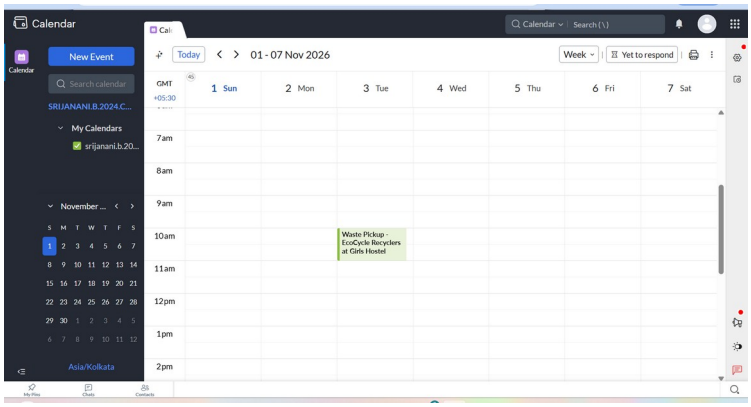


(c) Pickup approval e-mail confirming the approved waste, vendor, date and remarks

Figure 8.2: Scenario Step 2 – Confirmation and Alert E-mails



(a) Approval e-mail received after the Sustainability Officer set Status = Approved (vendor, pickup date 03-Nov-2026 and remarks)



(b) Resulting pickup event created in Zoho Calendar for 03 Nov 2026

Figure 8.3: Scenario Step 3 – Approval and Zoho Calendar Event

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Structured data model and sectioned form	Sections 4.2, 4.3, 5.2	One Waste Entry form with 20 fields in six sections
Automated calculation and rule enforcement	Sections 5.5, 5.6	Ten Deluge workflows; TC-04 to TC-12 passed
Reports and dashboard	Sections 5.3, 5.4	List report, kanban board and six-chart dashboard
Zoho Calendar integration and e-mail	Section 5.7	Calendar event created on approval; e-mails received
Role-based access	Sections 4.6, 5.8	Four profiles, PII flags and server-side role check
Testing and deployment readiness	Chapters 6, 7	18 test cases; 7 defects found and closed

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Recording a collection	Register entry, later copied to a spreadsheet	One form entry with defaults	No double entry
Error checking	None	Validation rules on every submission	Invalid and unsafe entries blocked
Pickup handling	Phone call; no record	Request, approval, e-mail and calendar event	Traceable and scheduled
Reporting	Monthly manual totals	Live dashboard and report	Real-time visibility
Data protection	Open spreadsheet	Profiles and PII flags	Need-to-know access

8.4 Limitations of the Low-Code Approach

Several platform constraints were met. A .ds file cannot carry users, role hierarchies, permission settings or connection authorisations, so these must be configured after import. A script cannot reference a connection that does not yet exist, so the calendar calls had to be activated after the connection was created. Some built-in Deluge tasks, such as zoho.cliq.postToUser and zoho.calendar.createEvent, did not behave as expected, and the direct REST API was used instead. The dashboard is limited to the chart types and layout options of the page builder, and the trial edition does not provide separate stage and production environments.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of the application is no-code: the form, its fields and sections, the list and kanban reports, the dashboard charts, the menu and the permission profiles were configured visually. The workflows are low-code: Deluge scripts express the calculations, validations, role checks and notifications in a few lines each. The calendar integration sits at the boundary with pro-code, as it calls a REST API with JSON parameters, but it still runs inside Creator through a managed connection. No part of the application required full pro-code development.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Green Campus Waste and Recycling Tracker replaces paper registers and spreadsheets with a single Zoho Creator application. One sectioned form captures every collection, ten Deluge workflows calculate its environmental impact, enforce waste-handling and role rules and send notifications, an approved pickup is scheduled in Zoho Calendar through its REST API, and a list report, a kanban board and a six-chart dashboard give the sustainability team an immediate picture of the campus.

The project shows that a useful, secure and maintainable business application can be built quickly on a low-code platform while still covering data modelling, reporting, automation, integration and role-based access.

9.2 Future Enhancements

- Promote locations, categories and vendors to master forms with lookup fields so the team can manage them without editing scripts.
- Add a scheduled workflow that e-mails a weekly summary of waste collected, recycled and CO2 saved.
- Use Zia to forecast waste volumes by location and season.
- Publish a student portal with a Green Points leaderboard to encourage segregation.
- Connect weighing scales or smart bins through the Creator REST API to record weights automatically.

9.3 Learning Outcomes and Career Relevance

This project developed skills in data modelling, form and report design, page building, Deluge scripting, workflow automation, REST API integration through OAuth connections, role-based security and systematic debugging. These skills apply directly to roles such as low-code developer, Zoho Creator consultant, business analyst and citizen developer, and the habit of tracing errors with logs and fixing them step by step applies to software development in general.

REFERENCES

[1] W. Rosam and O. Kalnichevski, Low-Code Development: A Practical Guide. New Delhi, India: BPB Publications.

[2] Zoho Corporation, “Zoho Creator Help Documentation.” [Online]. Available: https://www.zoho.com/creator/help/

[3] Zoho Corporation, “Deluge Script Reference Guide.” [Online]. Available: https://www.zoho.com/deluge/help/

[4] Zoho Corporation, “Zoho Calendar API Documentation.” [Online]. Available: https://www.zoho.com/calendar/help/api/

[5] Ministry of Environment, Forest and Climate Change, Government of India, “Solid Waste Management Rules,” 2016.

[6] Ministry of Environment, Forest and Climate Change, Government of India, “E-Waste (Management) Rules,” 2022.

[7] A. Sahay, A. Indamutsa, D. Di Ruscio and A. Pierantonio, “Supporting the understanding and comparison of low-code development platforms,” in Proc. 46th Euromicro Conf. Software Engineering and Advanced Applications (SEAA), 2020, pp. 171–178.

[8] A. C. Bock and U. Frank, “Low-code platform,” Business & Information Systems Engineering, vol. 63, no. 6, pp. 733–740, 2021.

[9] D. Di Ruscio, D. Kolovos, J. de Lara, A. Pierantonio, M. Tisi and M. Wimmer, “Low-code development and model-driven engineering: Two sides of the same coin?,” Software and Systems Modeling, vol. 21, pp. 437–446, 2022.

[10] S. Kaza, L. Yao, P. Bhada-Tata and F. Van Woerden, What a Waste 2.0: A Global Snapshot of Solid Waste Management to 2050. Washington, DC, USA: World Bank, 2018.

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Green_Campus_Waste_Recycling; Waste_Entry and dashboard sections	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Y	Waste Entry: Date, Radio, Drop Down, Decimal, Number, Multi Line, Single Line, Email	5.2	Table 4.2
II	Lookup field (relationship between forms)	N	Reference data kept as drop-downs to keep one form (Sec. 4.2)	4.2, 5.2	—
II	Subform (parent–child data)	N	Not needed: one record per collection	4.2, 5.2	—
II	Field validations / layout rules	Y	Mandatory fields; DS-08 validations; show/hide rules	5.2	Code 5.1
II	List report with filter, grouping, custom action	Y	Waste Entry Report – sort, search, filter, record actions	5.3	Table 5.1
II	Kanban / Calendar / Timeline / Map report	Y	waste_board grouped by Waste Category	5.3	Fig. 5.4
II	Pivot table / chart report	Y	Six charts on the dashboard page	5.3	Fig. 5.5
II	Page / dashboard built with UI Builder	Y	Green Campus Dashboard	5.4	Fig. 5.5
II	Deluge – form events (validate / success / field actions)	Y	On load, on user input, on validate, on success	5.5	Table 5.2
II	Deluge – functions, fetch, loops, collections	Y	Maps, conditions, string building, collections (calendar list)	5.5	App. C

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Email / in-app notifications	Y	Confirmation, admin alert and approval e-mails	5.5, 5.6	Code 5.2
II	Form workflow	Y	10 form workflows	5.6	Table 5.3
II	Scheduled workflow	N	Planned: weekly summary (Sec. 9.2)	5.6	—
II	Approval process	N	Approval handled through Status and role check	5.6	—
II	Blueprint	N	Not required	5.6	—
II	Batch / report workflow	N	Not required	5.6	—
II	Testing & debugging evidence	Y	18 test cases, 7 defects	6	Tables 6.1, 6.2
III	Integration with a Zoho service	Y	Zoho Calendar; Zoho Mail	5.7	Table 5.4
III	Connection / REST API call (invokeurl)	Y	zoho_calendar_connection; invokeurl GET/POST	5.7	Code 5.3
III	Webhook or Creator REST API	N	Not required	5.7	—
III	Users, roles and permission profiles	Y	Four permission profiles and role hierarchy	5.8	Fig. 5.8
III	Record-level access criteria / customer portal	N	PII flags and role checks used instead	5.8	—
III	Zia / AI feature	N	Future enhancement	5.9	—
III	Analytics (charts / Zoho Analytics)	Y	Dashboard charts	5.9	Fig. 5.5
IV	Performance optimisation applied & measured	Y	Maps instead of fetches; single-form design	7.2	Table 7.2
IV	Deployment across environments	Y	Development, Stage, Production (v1.0)	7.1	Fig. 7.1
IV	Security controls (encryption, OAuth, permissions)	Y	Profiles, PII, OAuth connection, server-side check	7.3	Table 7.3

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
IV	Governance (audit trail, administration)	Y	Record history, naming, .ds backup	7.4	Fig. 7.2
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Not required (Sec. 5.10)	5.10	—
V	Low-code limitations & spectrum analysis	Y	Sections 8.4, 8.5	8.4, 8.5	—
—	Mobile app / portal access	Y	Zoho Creator mobile app	5.11	Fig. 5.10

Total features implemented: 21 / 32

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	Good	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	Excellent	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	Good	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	Good	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	Satisfactory	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Waste_Entry form – On Load and On User Input workflows (DS-01 to DS-07)

```
// Script ID : DS-01
// Workflow  : 1. On Load - Defaults and Access
// Location  : Waste_Entry → on add → on load
input.Log_Date = zoho.currentdate;
input.Logged_By_Email = zoho.loginuserid;
input.Logged_By_Name = zoho.loginuser;
input.Status = "Logged";
input.Pickup_Required = "No";
hide Hazard_Notes;
hide Vendor;
hide Preferred_Pickup_Date;
hide Estimated_Value;
disable Carbon_Saved_Kg;
disable Green_Points;
disable Estimated_Value;
// RBAC: only the app owner (Administrator / Sustainability Officer) can
approve
if(zoho.loginuserid != zoho.adminuserid)
{
    disable Status;
    disable Approver_Remarks;
}

// Script ID : DS-02
// Workflow  : 2. On Edit Load - Access
// Location  : Waste_Entry → on edit → on load
disable Carbon_Saved_Kg;
disable Green_Points;
disable Estimated_Value;
if(zoho.loginuserid != zoho.adminuserid)
{
    disable Status;
    disable Approver_Remarks;
}
if(input.Pickup_Required == "Yes")
{
    show Vendor;
    show Preferred_Pickup_Date;
    show Estimated_Value;
}
else
{
    hide Vendor;
    hide Preferred_Pickup_Date;
    hide Estimated_Value;
}
if(input.Waste_Category == "Hazardous" || input.Waste_Category == "E-
Waste")
{
    show Hazard_Notes;
}
else
{
    hide Hazard_Notes;
}

// Script ID : DS-03
// Workflow  : 3. User Input - Category
// Location  : Waste_Entry → on add or edit → on user input of
Waste_Category
if(input.Waste_Category == "Hazardous" || input.Waste_Category == "E-
```

```
Waste")
{
    show Hazard_Notes;
}
else
{
    hide Hazard_Notes;
}
// ---- Auto-calculate CO2 saved, Green Points and pickup value ----
factorMap = Map();
factorMap.put("Plastic",1.5);
factorMap.put("Paper and Cardboard",0.9);
factorMap.put("Metal",4.0);
factorMap.put("Glass",0.3);
factorMap.put("E-Waste",2.0);
factorMap.put("Organic or Food",0.5);
factorMap.put("Hazardous",0.0);
factorMap.put("Mixed or General",0.0);
rateMap = Map();
rateMap.put("EcoCycle Recyclers",12.5);
rateMap.put("GreenTech E-Waste Solutions",35.0);
rateMap.put("Chennai Compost Co",2.0);
rateMap.put("ClearGlass Traders",4.0);
rateMap.put("PaperBack Mills",8.0);
factor = 0.0;
if(input.Waste_Category != null &&
factorMap.containKey(input.Waste_Category))
{
    factor = factorMap.get(input.Waste_Category).toDecimal();
}
w = ifnull(input.Weight_Kg,0.0);
pts = 0;
if(input.Disposal_Method == "Landfill")
{
    input.Carbon_Saved_Kg = 0.0;
    pts = (w * 2).round(0).toLong();
}
else
{
    input.Carbon_Saved_Kg = (w * factor).round(2);
    pts = (w * 10).round(0).toLong();
}
if(input.Segregated_Properly == "Yes")
{
    pts = pts + 5;
}
input.Green_Points = pts;
if(input.Pickup_Required == "Yes" && input.Vendor != null &&
rateMap.containKey(input.Vendor))
{
    input.Estimated_Value = (w *
rateMap.get(input.Vendor).toDecimal()).round(2);
}
else
{
    input.Estimated_Value = 0.0;
}

// Script ID : DS-04
// Workflow  : 4. User Input - Weight
// Location  : Waste_Entry → on add or edit → on user input of Weight_Kg
// ---- Auto-calculate CO2 saved, Green Points and pickup value ----
factorMap = Map();
factorMap.put("Plastic",1.5);
factorMap.put("Paper and Cardboard",0.9);
factorMap.put("Metal",4.0);
factorMap.put("Glass",0.3);
```

```
factorMap.put("E-Waste",2.0);
factorMap.put("Organic or Food",0.5);
factorMap.put("Hazardous",0.0);
factorMap.put("Mixed or General",0.0);
rateMap = Map();
rateMap.put("EcoCycle Recyclers",12.5);
rateMap.put("GreenTech E-Waste Solutions",35.0);
rateMap.put("Chennai Compost Co",2.0);
rateMap.put("ClearGlass Traders",4.0);
rateMap.put("PaperBack Mills",8.0);
factor = 0.0;
if(input.Waste_Category != null &&
factorMap.containKey(input.Waste_Category))
{
    factor = factorMap.get(input.Waste_Category).toDecimal();
}
w = ifnull(input.Weight_Kg,0.0);
pts = 0;
if(input.Disposal_Method == "Landfill")
{
    input.Carbon_Saved_Kg = 0.0;
    pts = (w * 2).round(0).toLong();
}
else
{
    input.Carbon_Saved_Kg = (w * factor).round(2);
    pts = (w * 10).round(0).toLong();
}
if(input.Segregated_Properly == "Yes")
{
    pts = pts + 5;
}
input.Green_Points = pts;
if(input.Pickup_Required == "Yes" && input.Vendor != null &&
rateMap.containKey(input.Vendor))
{
    input.Estimated_Value = (w *
rateMap.get(input.Vendor).toDecimal()).round(2);
}
else
{
    input.Estimated_Value = 0.0;
}

// Script ID : DS-05
// Workflow  : 5. User Input - Disposal
// Location  : Waste_Entry → on add or edit → on user input of
Disposal_Method
// ---- Auto-calculate CO2 saved, Green Points and pickup value ----
factorMap = Map();
factorMap.put("Plastic",1.5);
factorMap.put("Paper and Cardboard",0.9);
factorMap.put("Metal",4.0);
factorMap.put("Glass",0.3);
factorMap.put("E-Waste",2.0);
factorMap.put("Organic or Food",0.5);
factorMap.put("Hazardous",0.0);
factorMap.put("Mixed or General",0.0);
rateMap = Map();
rateMap.put("EcoCycle Recyclers",12.5);
rateMap.put("GreenTech E-Waste Solutions",35.0);
rateMap.put("Chennai Compost Co",2.0);
rateMap.put("ClearGlass Traders",4.0);
rateMap.put("PaperBack Mills",8.0);
factor = 0.0;
if(input.Waste_Category != null &&
factorMap.containKey(input.Waste_Category))
```

```
{
    factor = factorMap.get(input.Waste_Category).toDecimal();
}
w = ifnull(input.Weight_Kg,0.0);
pts = 0;
if(input.Disposal_Method == "Landfill")
{
    input.Carbon_Saved_Kg = 0.0;
    pts = (w * 2).round(0).toLong();
}
else
{
    input.Carbon_Saved_Kg = (w * factor).round(2);
    pts = (w * 10).round(0).toLong();
}
if(input.Segregated_Properly == "Yes")
{
    pts = pts + 5;
}
input.Green_Points = pts;
if(input.Pickup_Required == "Yes" && input.Vendor != null &&
rateMap.containKey(input.Vendor))
{
    input.Estimated_Value = (w *
rateMap.get(input.Vendor).toDecimal()).round(2);
}
else
{
    input.Estimated_Value = 0.0;
}

// Script ID : DS-06
// Workflow  : 6. User Input - Pickup
// Location  : Waste_Entry → on add or edit → on user input of
Pickup_Required
if(input.Pickup_Required == "Yes")
{
    show Vendor;
    show Preferred_Pickup_Date;
    show Estimated_Value;
}
else
{
    hide Vendor;
    hide Preferred_Pickup_Date;
    hide Estimated_Value;
}
// ---- Auto-calculate CO2 saved, Green Points and pickup value ----
factorMap = Map();
factorMap.put("Plastic",1.5);
factorMap.put("Paper and Cardboard",0.9);
factorMap.put("Metal",4.0);
factorMap.put("Glass",0.3);
factorMap.put("E-Waste",2.0);
factorMap.put("Organic or Food",0.5);
factorMap.put("Hazardous",0.0);
factorMap.put("Mixed or General",0.0);
rateMap = Map();
rateMap.put("EcoCycle Recyclers",12.5);
rateMap.put("GreenTech E-Waste Solutions",35.0);
rateMap.put("Chennai Compost Co",2.0);
rateMap.put("ClearGlass Traders",4.0);
rateMap.put("PaperBack Mills",8.0);
factor = 0.0;
if(input.Waste_Category != null &&
factorMap.containKey(input.Waste_Category))
{
```

```
    factor = factorMap.get(input.Waste_Category).toDecimal();
}
w = ifnull(input.Weight_Kg,0.0);
pts = 0;
if(input.Disposal_Method == "Landfill")
{
    input.Carbon_Saved_Kg = 0.0;
    pts = (w * 2).round(0).toLong();
}
else
{
    input.Carbon_Saved_Kg = (w * factor).round(2);
    pts = (w * 10).round(0).toLong();
}
if(input.Segregated_Properly == "Yes")
{
    pts = pts + 5;
}
input.Green_Points = pts;
if(input.Pickup_Required == "Yes" && input.Vendor != null &&
rateMap.containKey(input.Vendor))
{
    input.Estimated_Value = (w *
rateMap.get(input.Vendor).toDecimal()).round(2);
}
else
{
    input.Estimated_Value = 0.0;
}

// Script ID : DS-07
// Workflow  : 7. User Input - Vendor
// Location  : Waste_Entry → on add or edit → on user input of Vendor
// ---- Auto-calculate CO2 saved, Green Points and pickup value ----
factorMap = Map();
factorMap.put("Plastic",1.5);
factorMap.put("Paper and Cardboard",0.9);
factorMap.put("Metal",4.0);
factorMap.put("Glass",0.3);
factorMap.put("E-Waste",2.0);
factorMap.put("Organic or Food",0.5);
factorMap.put("Hazardous",0.0);
factorMap.put("Mixed or General",0.0);
rateMap = Map();
rateMap.put("EcoCycle Recyclers",12.5);
rateMap.put("GreenTech E-Waste Solutions",35.0);
rateMap.put("Chennai Compost Co",2.0);
rateMap.put("ClearGlass Traders",4.0);
rateMap.put("PaperBack Mills",8.0);
factor = 0.0;
if(input.Waste_Category != null &&
factorMap.containKey(input.Waste_Category))
{
    factor = factorMap.get(input.Waste_Category).toDecimal();
}
w = ifnull(input.Weight_Kg,0.0);
pts = 0;
if(input.Disposal_Method == "Landfill")
{
    input.Carbon_Saved_Kg = 0.0;
    pts = (w * 2).round(0).toLong();
}
else
{
    input.Carbon_Saved_Kg = (w * factor).round(2);
    pts = (w * 10).round(0).toLong();
}
```

```
if(input.Segregated_Properly == "Yes")
{
    pts = pts + 5;
}
input.Green_Points = pts;
if(input.Pickup_Required == "Yes" && input.Vendor != null &&
rateMap.containKey(input.Vendor))
{
    input.Estimated_Value = (w *
rateMap.get(input.Vendor).toDecimal()).round(2);
}
else
{
    input.Estimated_Value = 0.0;
}
```

C.2 Waste_Entry form – Validation and On Success workflows (DS-08 to DS-10)

```
// Script ID : DS-08
// Workflow  : 8. Validate and Calculate
// Location  : Waste_Entry → on add or edit → on validate
// ---- Business rule validations ----
if(input.Weight_Kg == null || input.Weight_Kg <= 0)
{
    alert "Weight must be greater than 0 kg.";
    cancel submit;
}
if(input.Weight_Kg > 1000)
{
    alert "More than 1000 kg in one entry looks wrong. Please re-check
the weight.";
    cancel submit;
}
if(input.Log_Date > zoho.currentdate)
{
    alert "Log date cannot be in the future.";
    cancel submit;
}
if((input.Waste_Category == "Hazardous" || input.Waste_Category == "E-
Waste") && (input.Hazard_Notes == null || input.Hazard_Notes == ""))
{
    alert "Please add handling notes for Hazardous or E-Waste.";
    cancel submit;
}
if(input.Waste_Category == "Hazardous" && input.Disposal_Method ==
"Landfill")
{
    alert "Hazardous waste cannot go to landfill. Choose Hazardous
Disposal.";
    cancel submit;
}
if(input.Pickup_Required == "Yes")
{
    if(input.Vendor == null || input.Vendor == "")
    {
        alert "Please choose a recycling vendor for the pickup.";
        cancel submit;
    }
    if(input.Preferred_Pickup_Date == null || input.Preferred_Pickup_Date
< zoho.currentdate)
    {
        alert "Please choose a pickup date that is today or later.";
        cancel submit;
    }
}
```

```
if(input.Status == null || input.Status == "")
{
    input.Status = "Logged";
}
if(input.Pickup_Required == "Yes" && input.Status == "Logged")
{
    input.Status = "Pickup Requested";
}
// ---- RBAC check on the server side ----
if(zoho.loginuserid != zoho.adminuserid && input.Status != "Logged" &&
input.Status != "Pickup Requested")
{
    alert "Only the Sustainability Officer (app admin) can approve or
close entries.";
    cancel submit;
}
// ---- Auto-calculate CO2 saved, Green Points and pickup value ----
factorMap = Map();
factorMap.put("Plastic",1.5);
factorMap.put("Paper and Cardboard",0.9);
factorMap.put("Metal",4.0);
factorMap.put("Glass",0.3);
factorMap.put("E-Waste",2.0);
factorMap.put("Organic or Food",0.5);
factorMap.put("Hazardous",0.0);
factorMap.put("Mixed or General",0.0);
rateMap = Map();
rateMap.put("EcoCycle Recyclers",12.5);
rateMap.put("GreenTech E-Waste Solutions",35.0);
rateMap.put("Chennai Compost Co",2.0);
rateMap.put("ClearGlass Traders",4.0);
rateMap.put("PaperBack Mills",8.0);
factor = 0.0;
if(input.Waste_Category != null &&
factorMap.containKey(input.Waste_Category))
{
    factor = factorMap.get(input.Waste_Category).toDecimal();
}
w = ifnull(input.Weight_Kg,0.0);
pts = 0;
if(input.Disposal_Method == "Landfill")
{
    input.Carbon_Saved_Kg = 0.0;
    pts = (w * 2).round(0).toLong();
}
else
{
    input.Carbon_Saved_Kg = (w * factor).round(2);
    pts = (w * 10).round(0).toLong();
}
if(input.Segregated_Properly == "Yes")
{
    pts = pts + 5;
}
input.Green_Points = pts;
if(input.Pickup_Required == "Yes" && input.Vendor != null &&
rateMap.containKey(input.Vendor))
{
    input.Estimated_Value = (w *
rateMap.get(input.Vendor).toDecimal()).round(2);
}
else
{
    input.Estimated_Value = 0.0;
}
info "Validated: " + input.Waste_Category + " " + input.Weight_Kg + " kg,
CO2 saved " + input.Carbon_Saved_Kg + " kg, points " +
```

```
input.Green_Points;

// Script ID : DS-09
// Workflow  : 9. On Success - Confirmation and Alerts
// Location  : Waste_Entry → on add → on success
pts = ifnull(input.Green_Points,0);
msg = "<p>Hi " + input.Logged_By_Name + ",</p><p>Your waste entry has
been recorded:</p><ul>";
msg = msg + "<li>Location: " + input.Campus_Location +
"</li><li>Category: " + input.Waste_Category + "</li>";
msg = msg + "<li>Weight: " + input.Weight_Kg + " kg</li><li>Disposal: " +
input.Disposal_Method + "</li>";
msg = msg + "<li>CO2 saved: " + input.Carbon_Saved_Kg + "
kg</li><li>Green Points: " + pts + "</li></ul>";
msg = msg + "<p>Keep our campus green!</p>";
sendmail
[
    from :zoho.loginuserid
    to :input.Logged_By_Email
    subject :"Waste logged - you earned " + pts + " Green Points"
    message :msg
]
info "Confirmation email sent to " + input.Logged_By_Email;
// Alert the admin for hazardous / bulk waste or a pickup request
if(input.Waste_Category == "Hazardous" || input.Weight_Kg >= 50 ||
input.Pickup_Required == "Yes")
{
    alertMsg = "<p>New entry needs attention:</p><ul><li>Location: " +
input.Campus_Location + "</li><li>Category: " + input.Waste_Category +
"</li><li>Weight: " + input.Weight_Kg + " kg</li><li>Pickup required: " +
input.Pickup_Required + "</li><li>Vendor: " + ifnull(input.Vendor,"-") +
"</li><li>Logged by: " + input.Logged_By_Name + "</li></ul>";
    sendmail
    [
        from :zoho.loginuserid
        to :zoho.adminuserid
        subject :"Green Campus alert: " + input.Waste_Category + " at " +
input.Campus_Location
        message :alertMsg
    ]
    info "Admin alert sent";
}

// Script ID : DS-10
// Workflow  : 10. On Approval - Notify and Zoho Calendar
// Location  : Waste_Entry → on edit → on success
if(input.Status == "Approved" && input.Pickup_Required == "Yes")
{
    note = "<p>Your pickup request for " + input.Weight_Kg + " kg of " +
input.Waste_Category + " at " + input.Campus_Location + " has been
<b>approved</b>.</p><p>Vendor: " + input.Vendor + "<br>Pickup date: " +
input.Preferred_Pickup_Date + "<br>Remarks: " +
ifnull(input.Approver_Remarks,"-") + "</p>";
    sendmail
    [
        from :zoho.loginuserid
        to :input.Logged_By_Email
        subject :"Pickup approved - " + input.Campus_Location
        message :note
    ]
    info "Approval email sent";
    // ===== INTEGRATION: Zoho Calendar REST API via connection
"zoho_calendar_connection" =====
    // Builds a 10:00-11:00 pickup event on the pickup date.
    datePart = input.Preferred_Pickup_Date.toString("yyyyMMdd");
    dateAndTime = Map();
    dateAndTime.put("timezone","Asia/Calcutta");
```

```
    dateAndTime.put("start",datePart + "T100000+0530");
    dateAndTime.put("end",datePart + "T110000+0530");
    eventMap = Map();
    eventMap.put("dateandtime",dateAndTime);
    eventMap.put("title","Waste Pickup - " + input.Vendor + " at " +
input.Campus_Location);
    eventMap.put("description",input.Waste_Category + " - " +
input.Weight_Kg + " kg");
    info "Event data: " + eventMap;
    // TO ACTIVATE: turn on Connection Access for this app, then remove
the // from the lines below
    // calendarId = "";
    // calResp = invokeurl
    // [
    //     url :"https://calendar.zoho.in/api/v1/calendars"
    //     type :GET
    //     connection:"zoho_calendar_connection"
    // ];
    // calList = calResp.get("calendars");
    // if(calList != null && calList.size() > 0)
    // {
    //     calendarId = calList.get(0).get("uid");
    // }
    // params = Map();
    // params.put("eventdata",eventMap.toString());
    // response = invokeurl
    // [
    //     url :"https://calendar.zoho.in/api/v1/calendars/" + calendarId
+ "/events"
    //     type :POST
    //     parameters:params
    //     connection:"zoho_calendar_connection"
    // ];
    // info response;
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the application link shared by the Sustainability Officer and sign in with your Zoho account, or install the Zoho Creator app on Android or iOS, sign in and open Green_Campus_Waste_Recycling. The left menu shows the Waste Entry form, the Waste Entry Report, the waste_board and the Green Campus Dashboard, according to your role.

D.2 Instructions for Department Coordinators, Waste Collectors and Student Volunteers

- Open Waste Entry. The date, your name and e-mail are already filled.
- Choose the shift, campus location, department and waste category, enter the weight and choose the disposal method and whether the waste was segregated. CO2 saved and Green Points appear automatically.
- For hazardous waste or e-waste, describe how it is packed in Hazard / Handling Notes.
- To send recyclables to a vendor, set Vendor Pickup Required to Yes, choose the vendor and a pickup date. The estimated value is shown.
- Click Submit. Correct any field named in an alert, then submit again. A confirmation e-mail is sent to you.
- Use the Waste Entry Report or the waste_board to find entries, and the Green Campus Dashboard to see campus totals.

D.3 Instructions for the Sustainability Officer (Administrator)

- Watch for alert e-mails about hazardous waste, large quantities and pickup requests.
- Open the entry from the report or the waste_board, set Status to Approved or Rejected and add remarks; the requester is e-mailed and approved pickups appear in Zoho Calendar.
- Set the status to Collected and then Recycled as the pickup is completed.
- Manage users, roles and permission profiles in Settings → User Permissions.
- Review the dashboard regularly and export the report when a summary is needed.